Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Github link :- https://github.com/kaladiyafarhan9879/Machine-learning-.git

Ouput :-

```
All libraries imported successfully.
```

+ Code + Markdown

```
Available files:
/kaggle/input/datasets/elemento/nyc-yellow-taxi-trip-data/yellow_tripdata_2016-01.csv
/kaggle/input/datasets/elemento/nyc-yellow-taxi-trip-data/yellow_tripdata_2016-03.csv
/kaggle/input/datasets/elemento/nyc-yellow-taxi-trip-data/yellow_tripdata_2016-02.csv
/kaggle/input/datasets/elemento/nyc-yellow-taxi-trip-data/yellow_tripdata_2015-01.csv

Selected dataset:
/kaggle/input/datasets/elemento/nyc-yellow-taxi-trip-data/yellow_tripdata_2016-01.csv

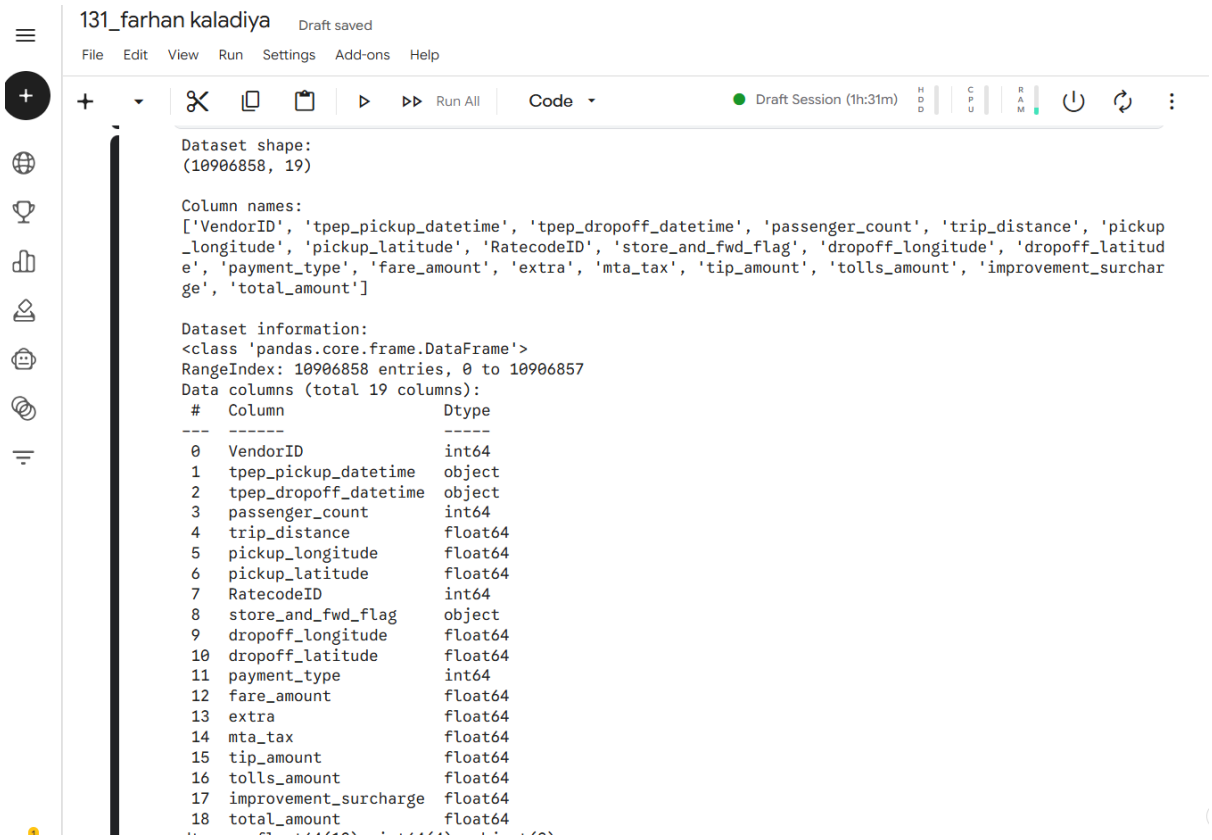
Dataset loaded successfully.
Dataset shape: (10906858, 19)

First 5 rows:
```

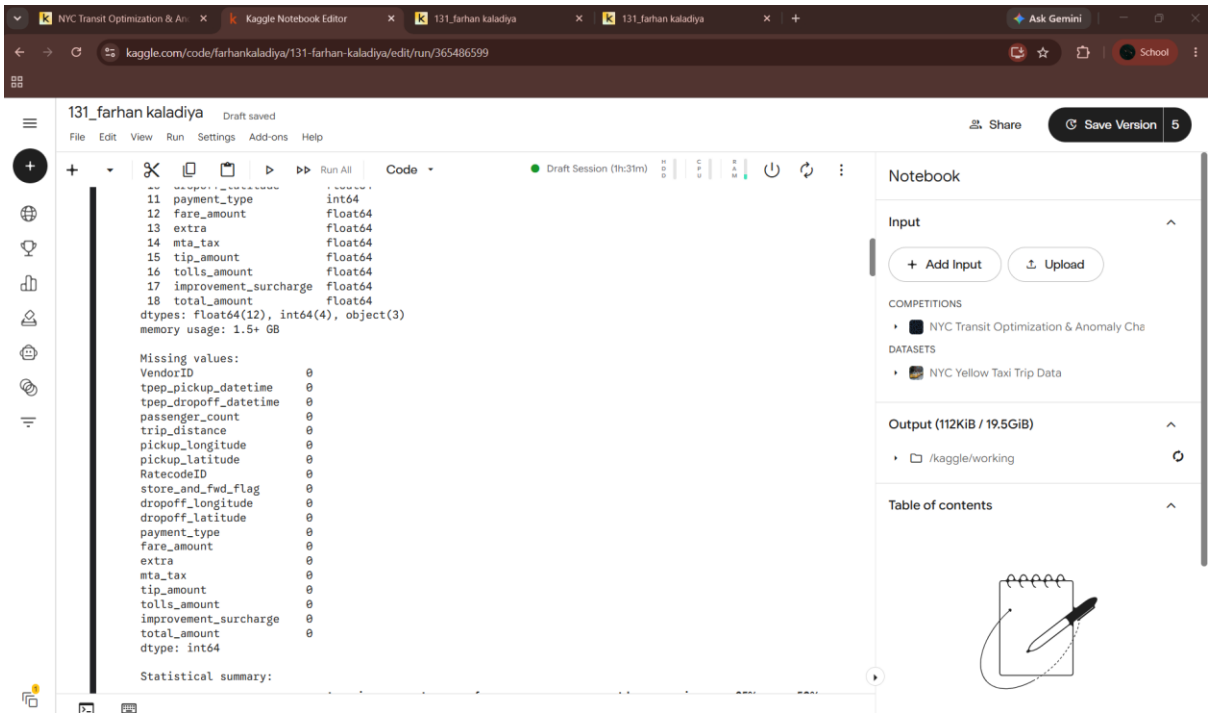
	VendorID	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	pickup_longitude	pickup_latitu
0	2	2016-01-01 00:00:00	2016-01-01 00:00:00	2	1.10	-73.990372	40.734
1	2	2016-01-01 00:00:00	2016-01-01 00:00:00	5	4.90	-73.980782	40.729
2	2	2016-01-01 00:00:00	2016-01-01 00:00:00	1	10.54	-73.984550	40.679
3	2	2016-01-01 00:00:00	2016-01-01 00:00:00	1	4.75	-73.993469	40.718
4	2	2016-01-01 00:00:00	2016-01-01 00:00:00	3	1.76	-73.960625	40.781

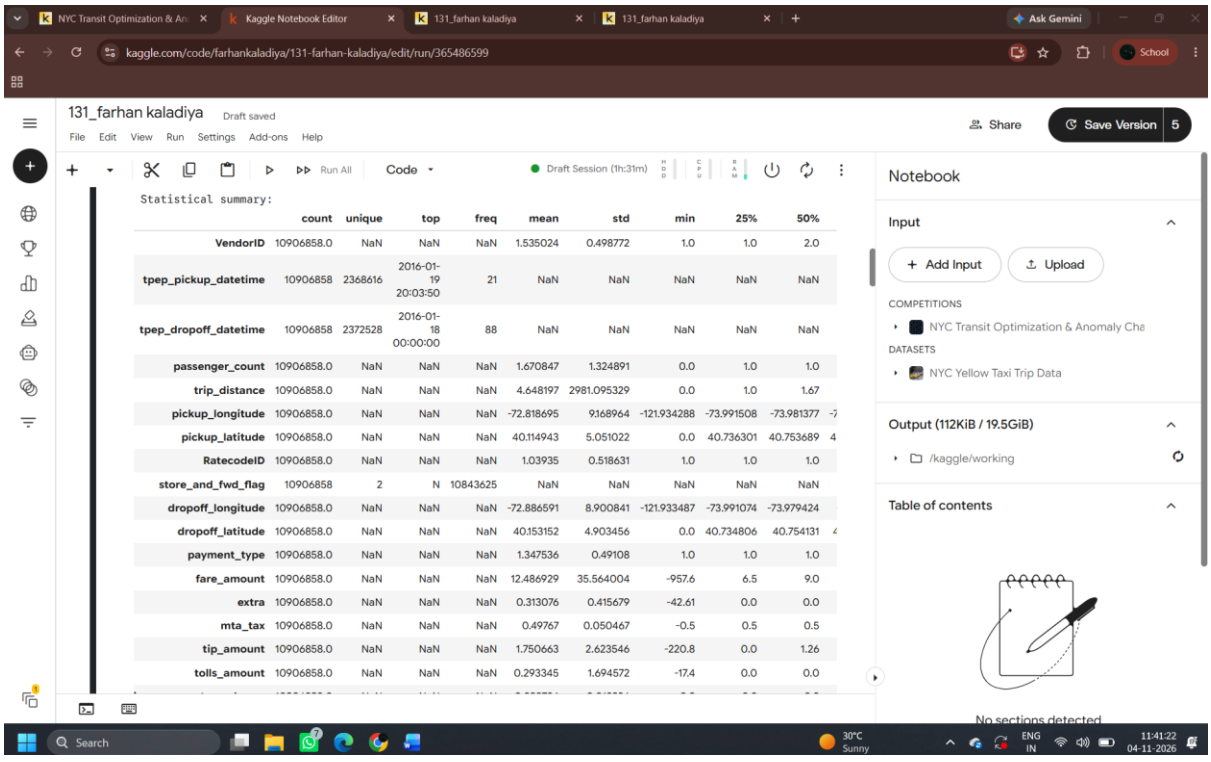
```
Dataset columns:
['VendorID', 'tpep_pickup_datetime', 'tpep_dropoff_datetime', 'passenger_count', 'trip_distance', 'pickup
_longitude', 'pickup_latitude', 'RatecodeID', 'store_and_fwd_flag', 'dropoff_longitude', 'dropoff_latitud
e', 'payment_type', 'fare_amount', 'extra', 'mta_tax', 'tip_amount', 'tolls_amount', 'improvement_surchar
ge', 'total_amount']
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**



Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**





STANDARDIZE COLUMN NAMES

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

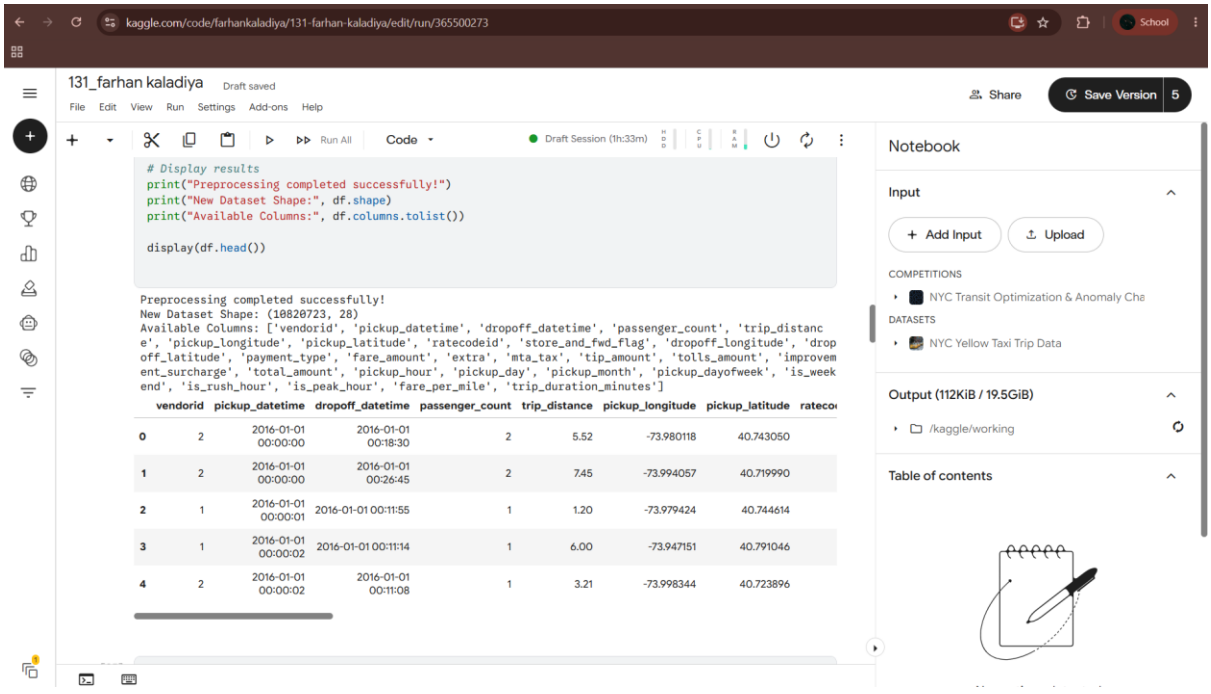
```
print("Updated column names:")

print(df.columns.tolist())
```

```
Updated column names:
['vendorid', 'pickup_datetime', 'dropoff_datetime', 'passenger_count', 'trip_distance', 'pickup_longitud
e', 'pickup_latitude', 'ratecodeid', 'store_and_fwd_flag', 'dropoff_longitude', 'dropoff_latitude', 'paym
ent_type', 'fare_amount', 'extra', 'mta_tax', 'tip_amount', 'tolls_amount', 'improvement_surcharge', 'tot
al_amount', 'pickup_hour', 'pickup_day', 'pickup_month', 'pickup_dayofweek', 'is_weekend', 'is_rush_hou
r', 'is_peak_hour', 'fare_per_mile']
```

+ Code + Markdown

DATA PREPROCESSING

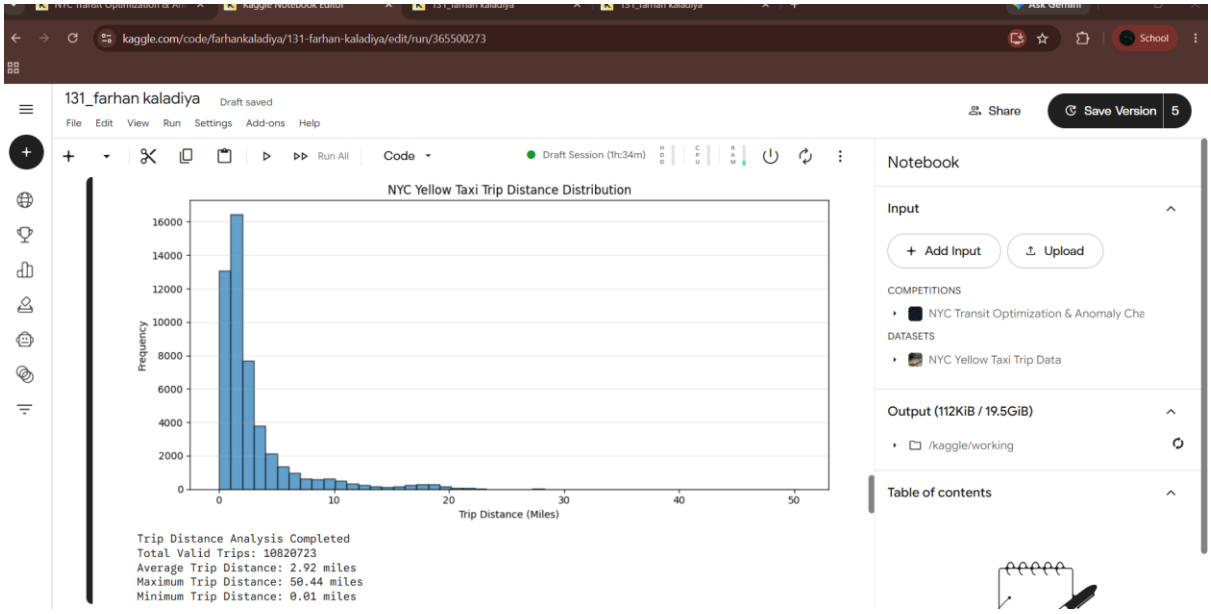



```
Missing columns:
['pulocationid', 'dolocationid']

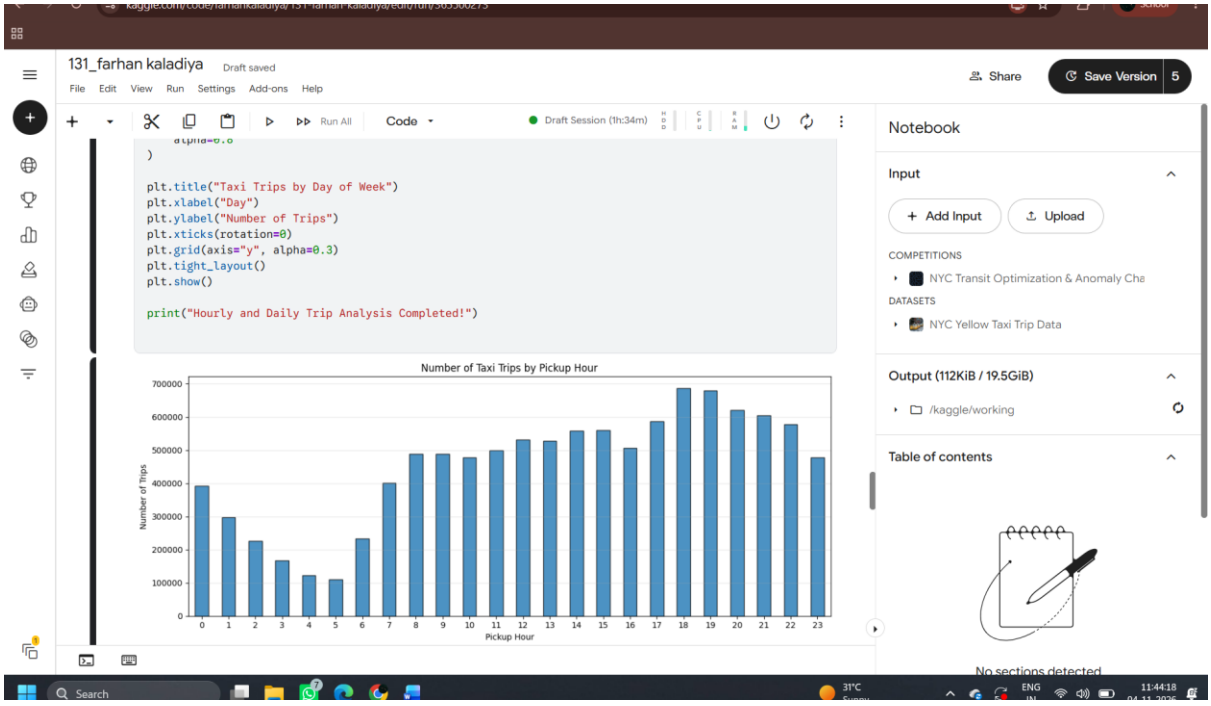
Available columns:
['vendorid', 'pickup_datetime', 'dropoff_datetime', 'passenger_count', 'trip_distance', 'pickup_longitud
e', 'pickup_latitude', 'ratecodeid', 'store_and_fwd_flag', 'dropoff_longitude', 'dropoff_latitude', 'paym
ent_type', 'fare_amount', 'extra', 'mta_tax', 'tip_amount', 'tolls_amount', 'improvement_surcharge', 'tot
al_amount', 'pickup_hour', 'pickup_day', 'pickup_month', 'pickup_dayofweek', 'is_weekend', 'is_rush_hou
r', 'is_peak_hour', 'fare_per_mile', 'trip_duration_minutes', 'average_speed_mph']
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

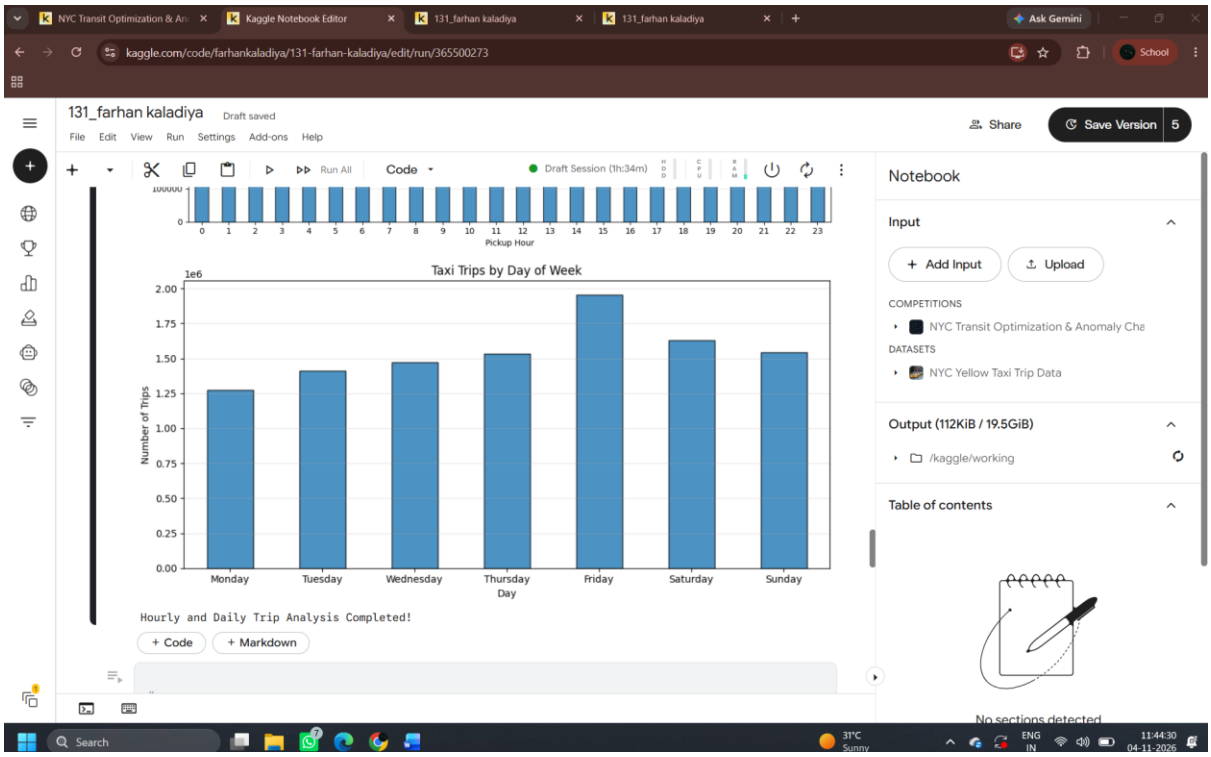
TRIP DISTANCE ANALYSIS.



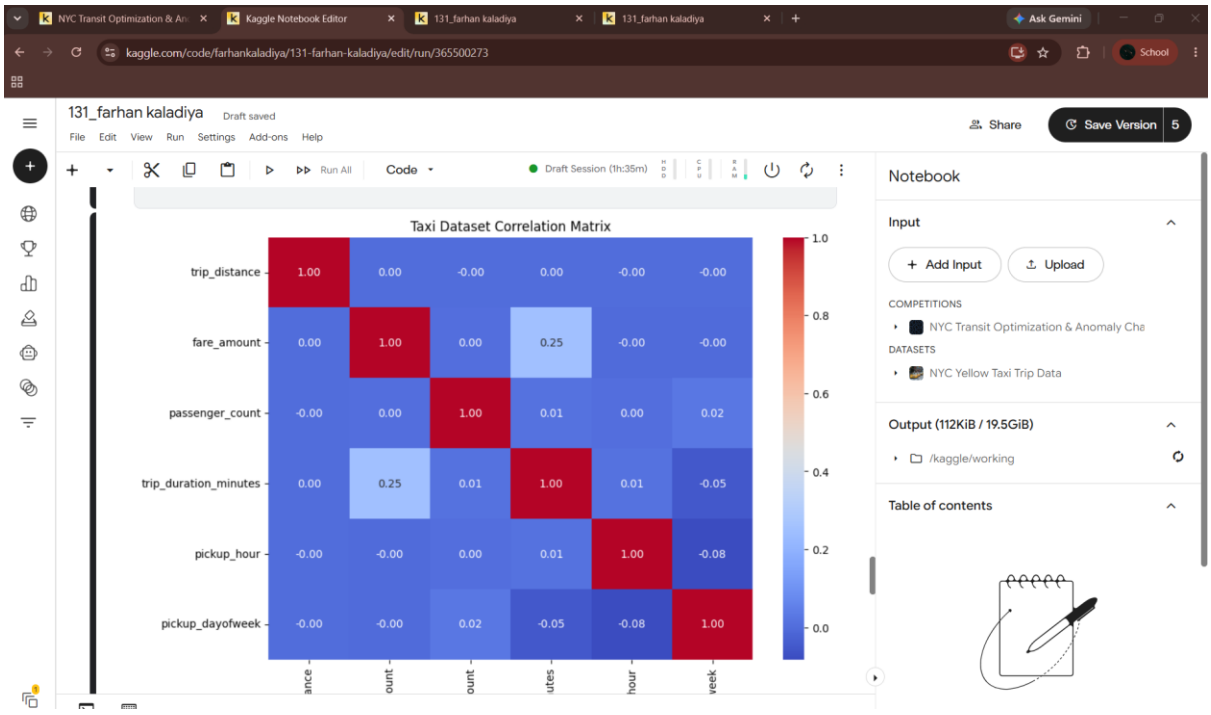
CREATE HOUR AND DAY COLUMNS



Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**



CORRELATION ANALYSIS



Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

PILLAR A TARGET

```
Pillar A target created.
Surge_Tier
0    4425391
1    3244503
2    3150829
Name: count, dtype: int64
```

+ Code + Markdown

```
import pandas as pd
import numpy as np
```

PILLAR B TARGET

.

```
            "CEPI_Score"
        ]
    ].head()
)
```

```
Pillar B target created successfully!
Dataset Shape: (10820723, 40)

CEPI Score Statistics:
count    1.082072e+07
mean     1.644891e-01
std      8.361051e-02
min      1.678755e-02
25%      9.255433e-02
50%      1.401004e-01
75%      2.235536e-01
max      7.097797e-01
Name: CEPI_Score, dtype: float64

Sample Results:
```

	zone_pair	passenger_penalty	idling_penalty	zone_topological_weight	CEPI_Score
0	40.743_-73.98_40.763_-73.913	0.250	0.039100	0.035170	0.101191
1	40.72_-73.994_40.79_-73.966	0.250	0.041890	0.035170	0.102307
2	40.745_-73.979_40.754_-73.992	0.125	0.115694	0.111485	0.117223
3	40.791_-73.947_40.866_-73.921	0.125	0.021778	0.035170	0.056762

PILLAR C FEATURES

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

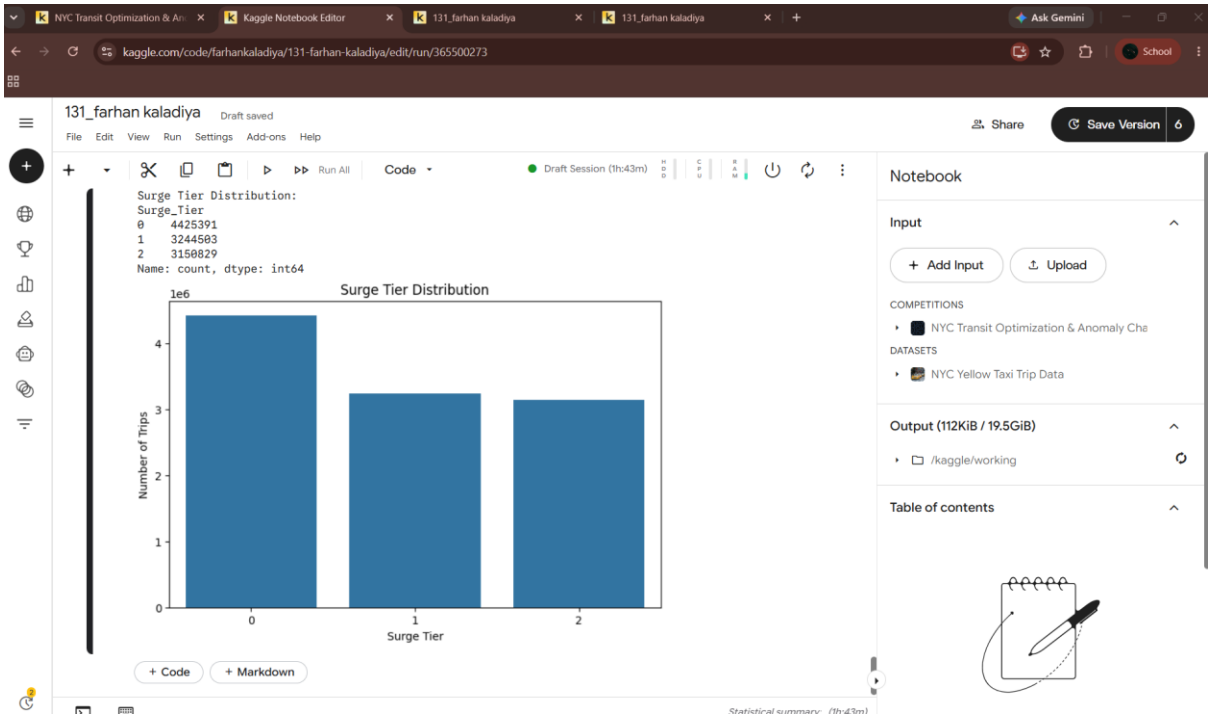
```
Missing columns: ['pulocationid', 'dolocationid']
Available columns: ['vendorid', 'pickup_datetime', 'dropoff_datetime', 'passenger_count', 'trip_distanc
e', 'pickup_longitude', 'pickup_latitude', 'ratecodeid', 'store_and_fwd_flag', 'dropoff_longitude', 'drop
off_latitude', 'payment_type', 'fare_amount', 'extra', 'mta_tax', 'tip_amount', 'tolls_amount', 'improvem
ent_surcharge', 'total_amount', 'pickup_hour', 'pickup_day', 'pickup_month', 'pickup_dayofweek', 'is_week
end', 'is_rush_hour', 'is_peak_hour', 'fare_per_mile', 'trip_duration_minutes', 'average_speed_mph', 'sur
ge_tier', 'passenger_penalty', 'duration_per_mile', 'duration_penalty', 'idling_penalty', 'zone_pair', 's
ame_zone', 'zone_pair_count', 'zone_frequency_normalized', 'zone_topological_weight', 'cepi_score']
```

+ Code + Markdown

Pillar A – Classification (Surge Tier Prediction)

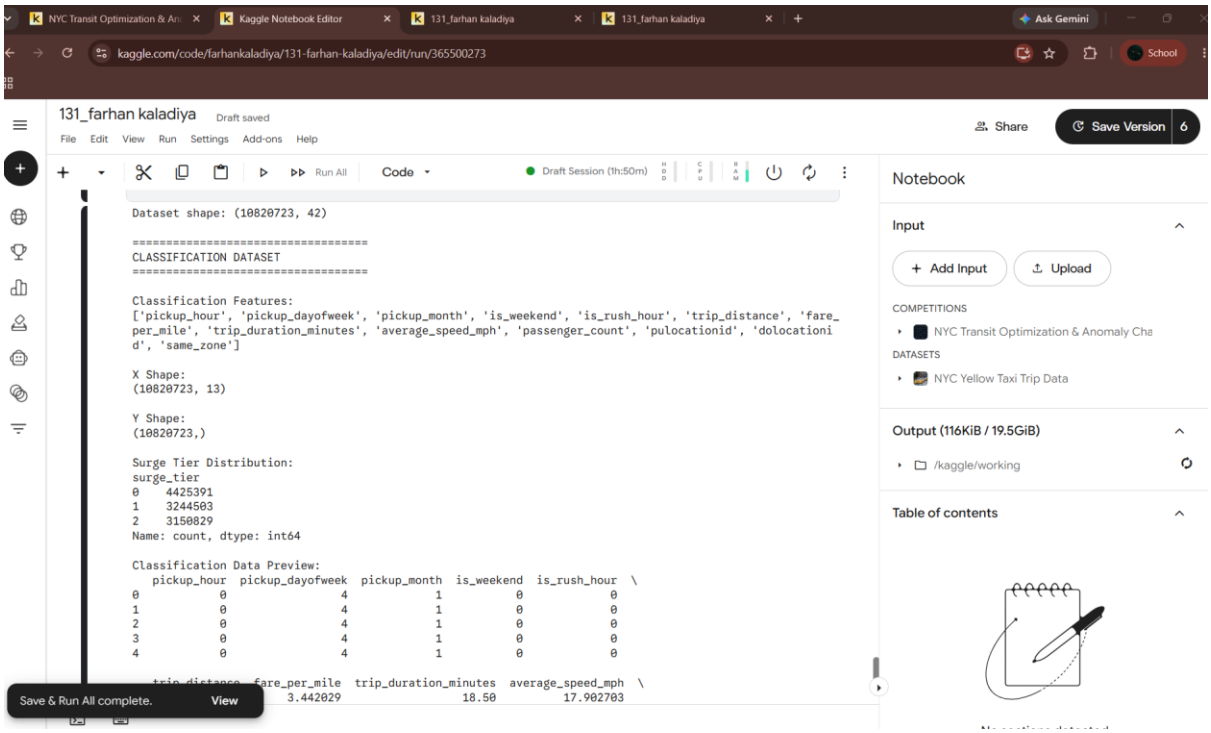
Target construction

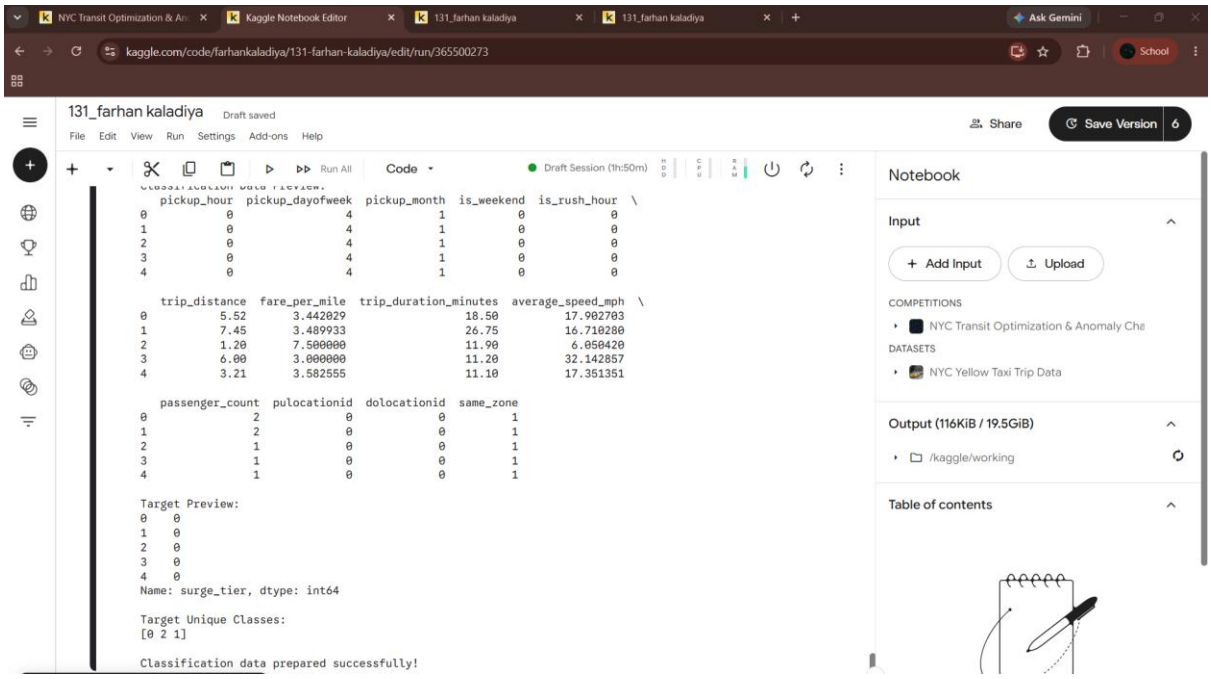
Output :-



CLASSIFICATION FEATURES

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**





Model development

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Output :-

```
Dataset shape: (10820723, 13)
Target shape: (10820723,)
Training data: (8656578, 13)
Testing data: (2164145, 13)
Training classification model...
Classification model trained successfully!

Classification Accuracy: 99.99 %

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    885078
           1       1.00      1.00      1.00    648901
           2       1.00      1.00      1.00    630166

    accuracy                           1.00   2164145
   macro avg       1.00      1.00      1.00   2164145
weighted avg       1.00      1.00      1.00   2164145

PART 9 COMPLETED SUCCESSFULLY!
```

+ Code + Markdown

Model Evaluation:-

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

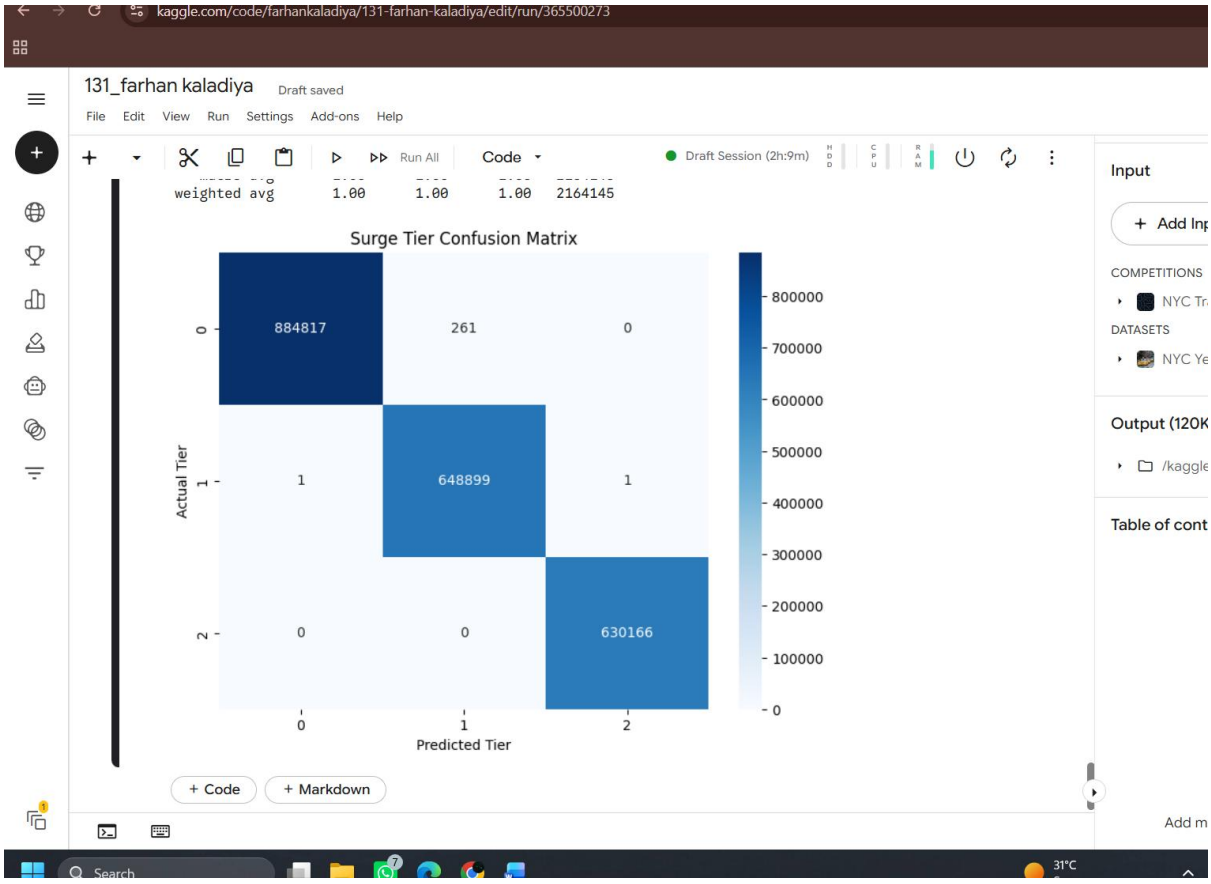
Output :-

```
Classification Results
----------------------
Accuracy: 0.999878473946986
Macro F1 Score: 0.9998828551231306

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00    885078
           1       1.00      1.00      1.00    648901
           2       1.00      1.00      1.00    630166

    accuracy                           1.00   2164145
   macro avg       1.00      1.00      1.00   2164145
weighted avg       1.00      1.00      1.00   2164145
```



Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Prediction generation

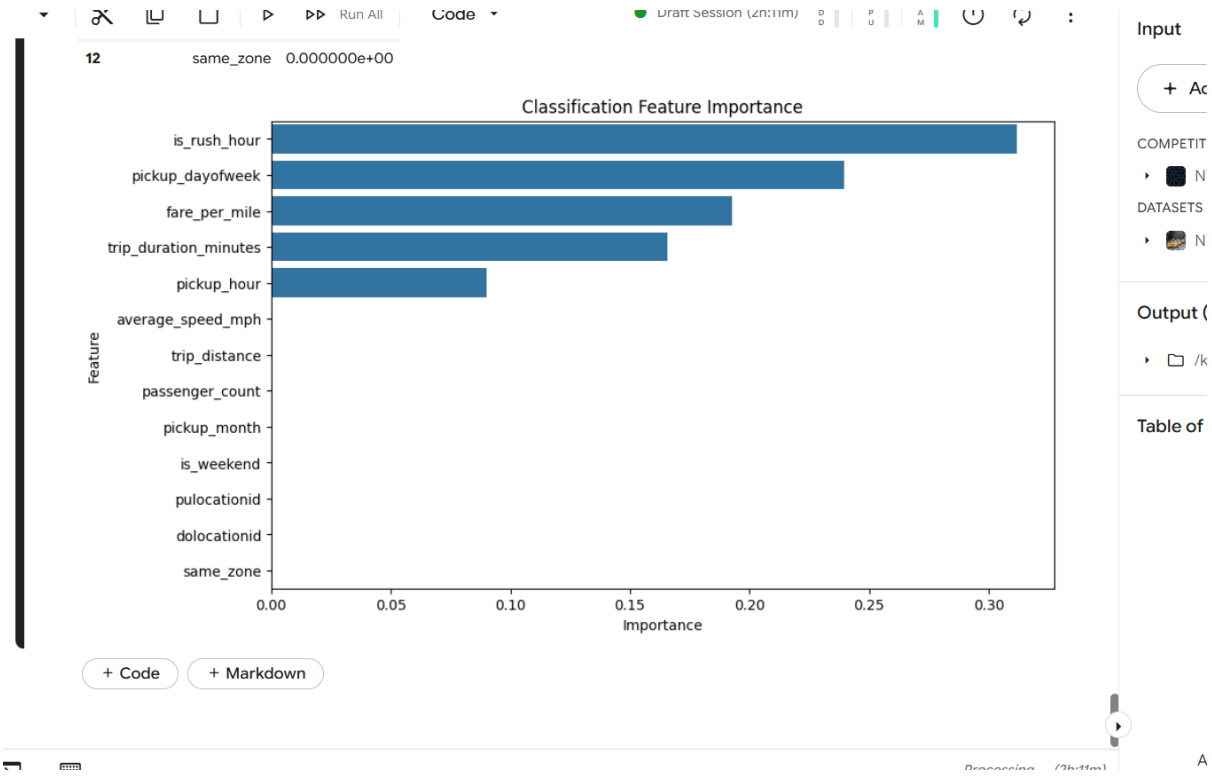
Output :-

```
plt.show()
```

Feature Importance:

	Feature	Importance
4	is_rush_hour	3.118365e-01
1	pickup_dayofweek	2.394532e-01
6	fare_per_mile	1.925616e-01
7	trip_duration_minutes	1.656761e-01
0	pickup_hour	9.009603e-02
8	average_speed_mph	3.764743e-04
5	trip_distance	7.001745e-08
9	passenger_count	1.053949e-20
2	pickup_month	0.000000e+00
3	is_weekend	0.000000e+00
10	pulocationid	0.000000e+00
11	dolocationid	0.000000e+00
12	same_zone	0.000000e+00

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**



Pillar B – Regression (CEPI Score Prediction)

Target construction

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Output :-

```
        [
            "passenger_count",
            "passenger_penalty"
        ]
    ].head()
)
```

	passenger_count	passenger_penalty
0	2	0.250
1	2	0.250
2	1	0.125
3	1	0.125
4	1	0.125

+ Code + Markdown

Feature preparation

Output :-

```
            "trip_distance",
            "duration_per_mile",
            "idling_penalty"
        ]
    ].head()
)
```

	trip_duration_minutes	trip_distance	duration_per_mile	idling_penalty
0	18.50	5.52	3.351449	0.039100
1	26.75	7.45	3.590604	0.041890
2	11.90	1.20	9.916667	0.115694
3	11.20	6.00	1.866667	0.021778
4	11.10	3.21	3.457944	0.040343

+ Code + Markdown

Model development

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Output :-

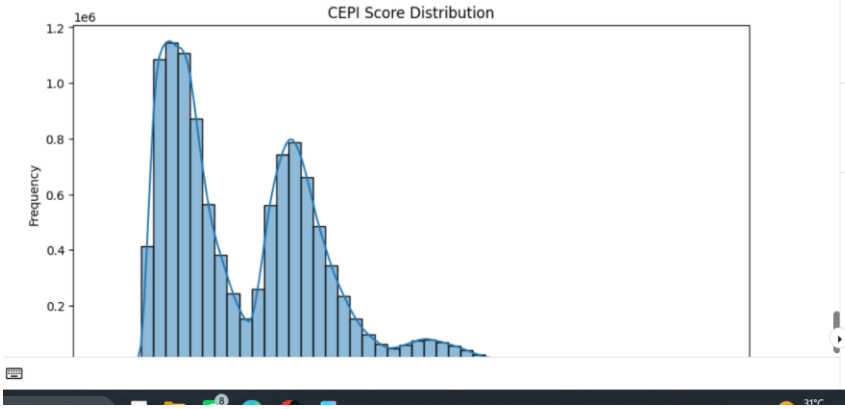
```
        ]
    ].head()
)
```

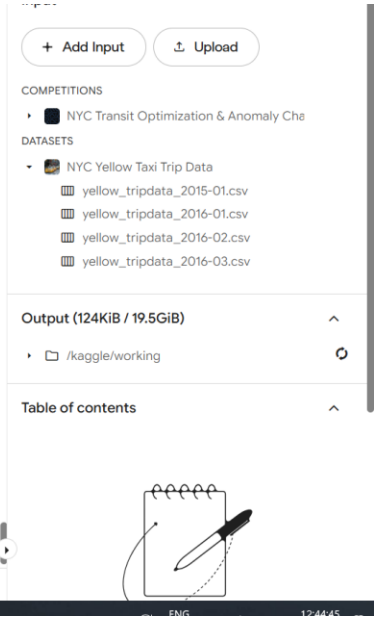
	zone_pair	zone_pair_count	zone_frequency_normalized	zone_topological_weight
0	40.743_-73.98_40.763_-73.913	1	0.058616	0.185170
1	40.72_-73.994_40.79_-73.966	1	0.058616	0.185170
2	40.745_-73.979_40.754_-73.992	8	0.185809	0.261485
3	40.791_-73.947_40.866_-73.921	1	0.058616	0.185170
4	40.724_-73.998_40.688_-73.996	2	0.092904	0.205743

+ Code + Markdown

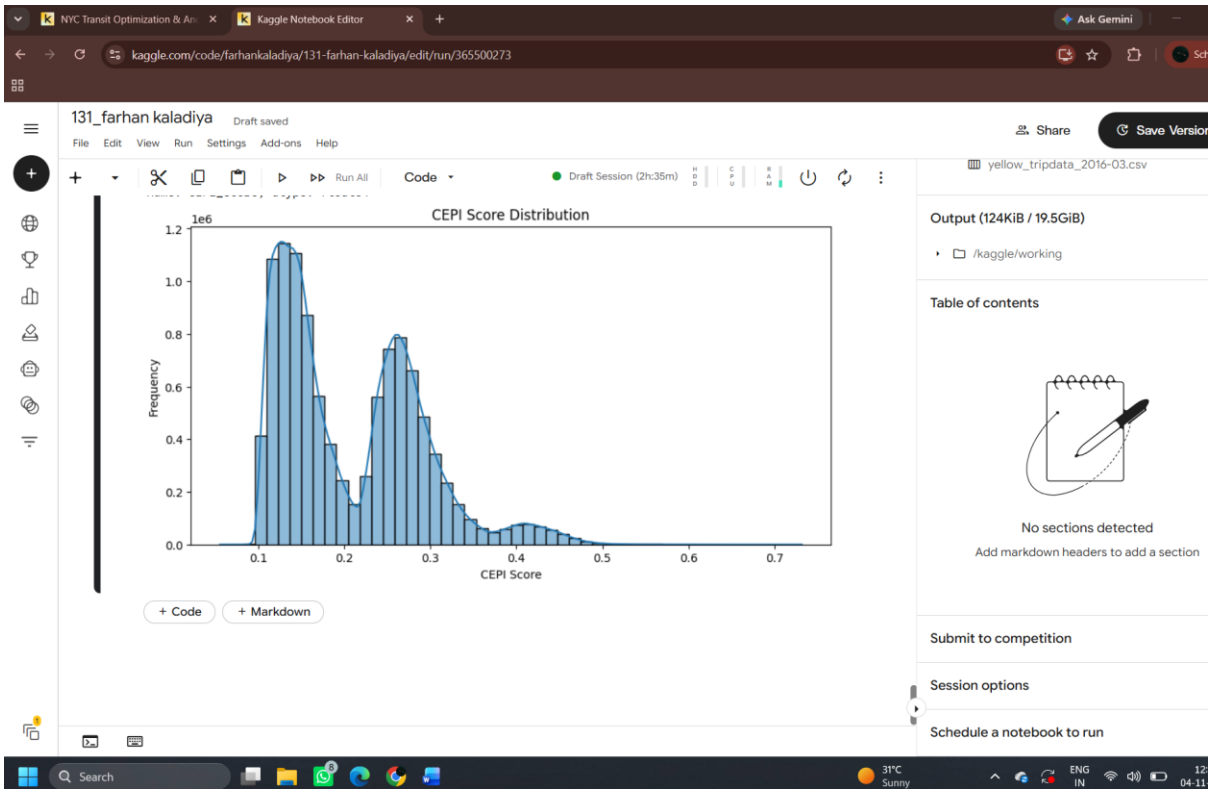
```
CEPI Score created.

CEPI Statistics:
count    1.082072e+07
mean     2.087103e-01
std      8.237527e-02
min      5.562164e-02
25%      1.374821e-01
50%      1.845665e-01
75%      2.684293e-01
max      7.318070e-01
Name: CEPI_Score, dtype: float64
```





Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**



Target construction:-

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Output :-

```
Regression X shape:
(10820723, 13)

Regression y shape:
(10820723,)
```

+ Code + Markdown

Feature preparation:-

Output :-

```
Dataset shape: (100000, 13)

Training regression model...
Regression model trained successfully!

REGRESSION MODEL RESULTS
------------------------
MAE: 0.0011417827695960592
MSE: 5.422701654070047e-06
RMSE: 0.0023286695029716103
R2 Score: 0.9991873288835771
```

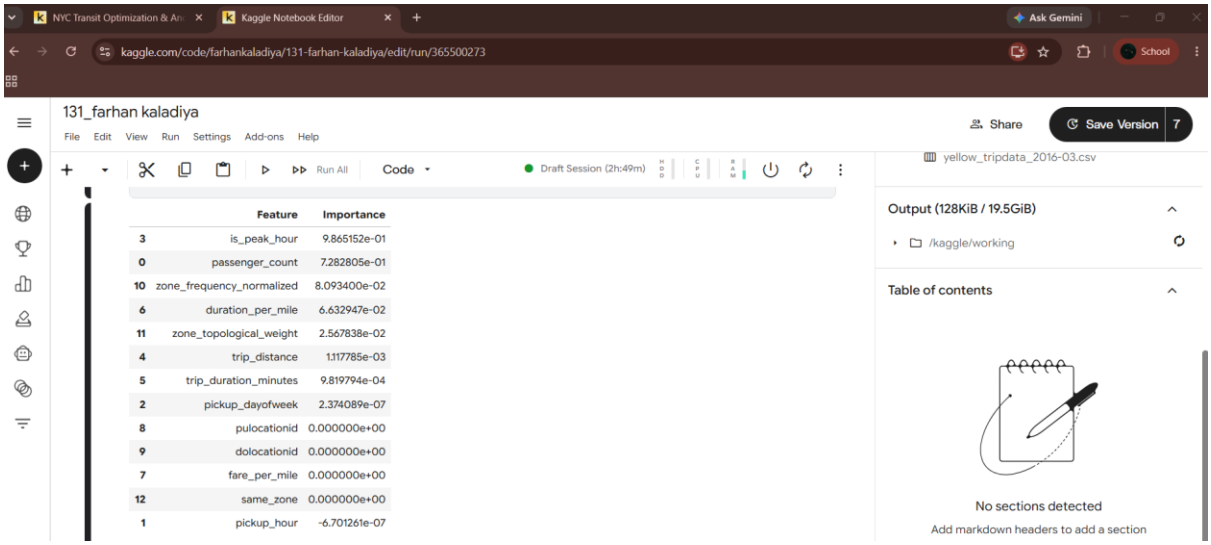
+ Code + Markdown

Model development

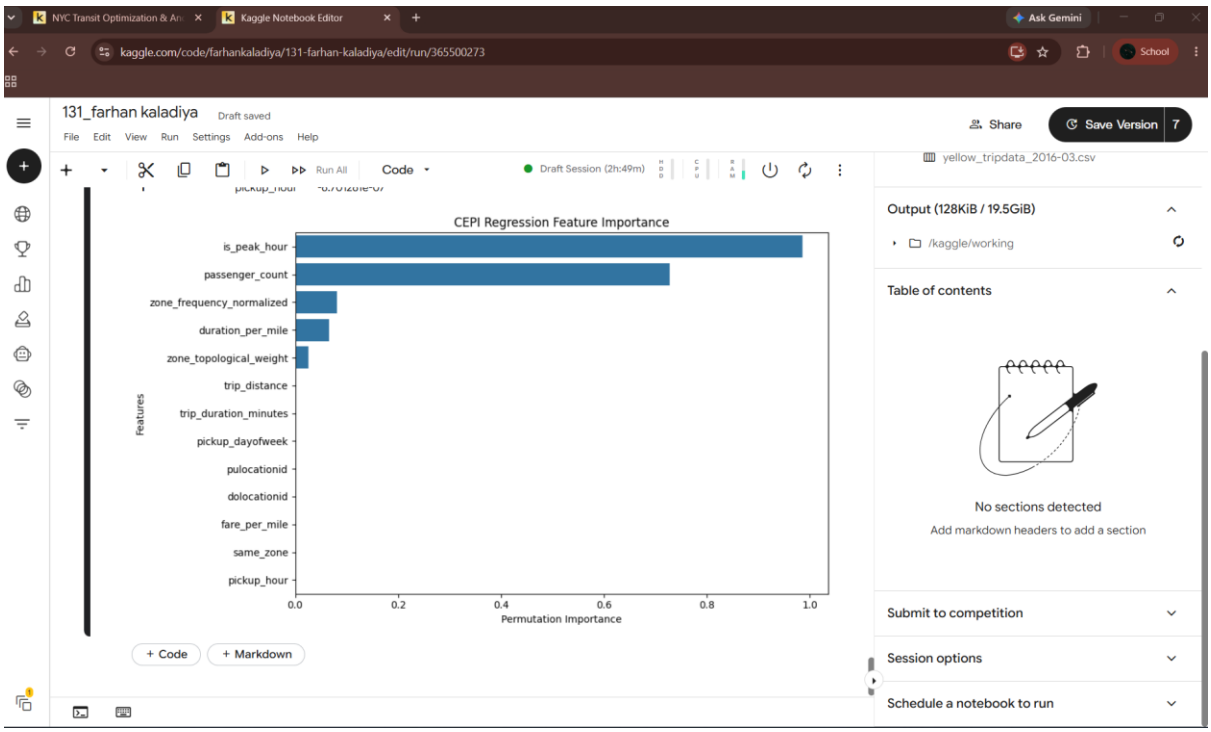
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Model evaluation

Prediction



	Feature	Importance
3	is_peak_hour	9.865152e-01
0	passenger_count	7.282805e-01
10	zone_frequency_normalized	8.093400e-02
6	duration_per_mile	6.632947e-02
11	zone_topological_weight	2.567838e-02
4	trip_distance	1.117785e-03
5	trip_duration_minutes	9.819794e-04
2	pickup_dayofweek	2.374089e-07
8	pulocationid	0.000000e+00
9	dolocationid	0.000000e+00
7	fare_per_mile	0.000000e+00
12	same_zone	0.000000e+00
1	pickup_hour	-6.701261e-07



Pillar C – Unsupervised Learning (Ghost Corridor Identification)

Zone understanding

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

```
Clustering preprocessing completed.
Clustering matrix shape:
(10820723, 9)
```

+ Code + Markdown

Feature engineering

Clustering

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

```
Clustering completed.

Cluster distribution:
Ghost_Corridor_Cluster
0    1219163
1    3665548
2    2487204
3    3311867
4     136941
Name: count, dtype: int64
```

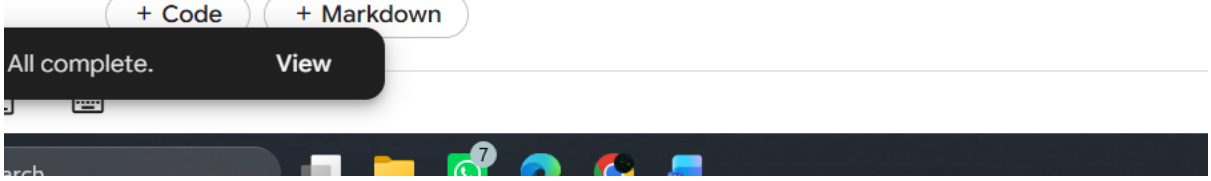
+ Code + Markdown



Cluster evaluation

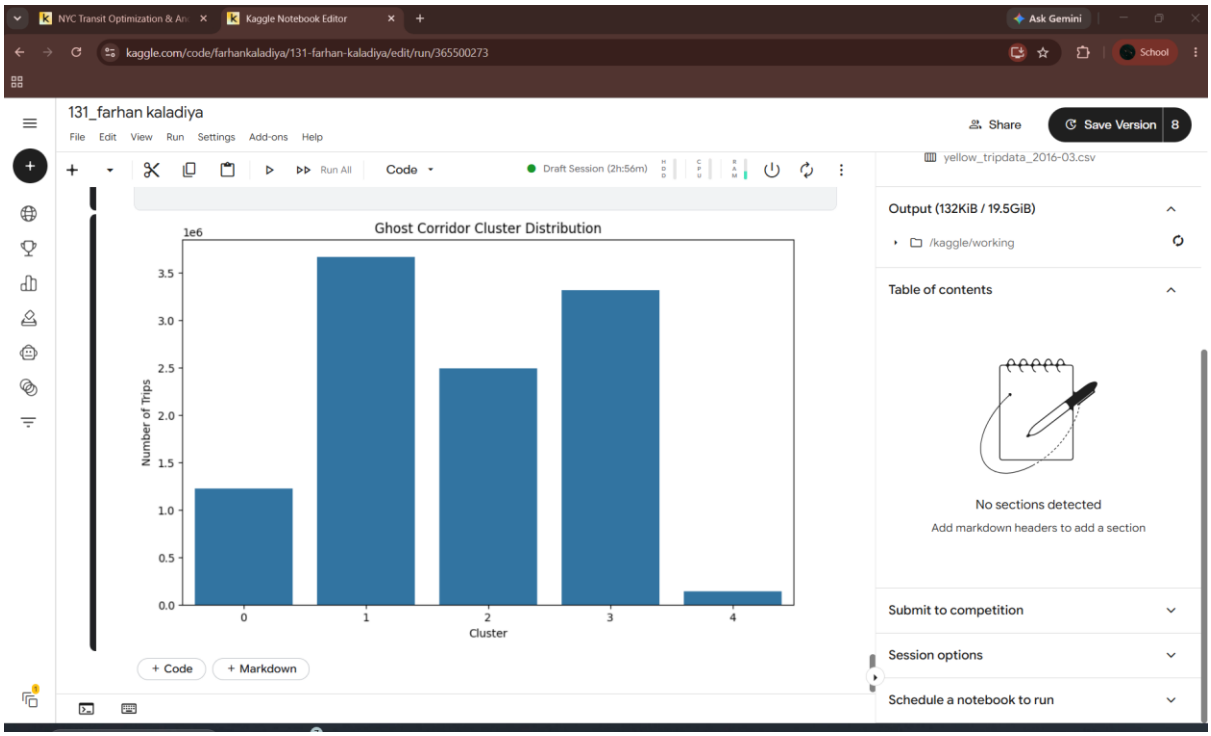
```
Sample Silhouette Score: 0.2634008894466617
```

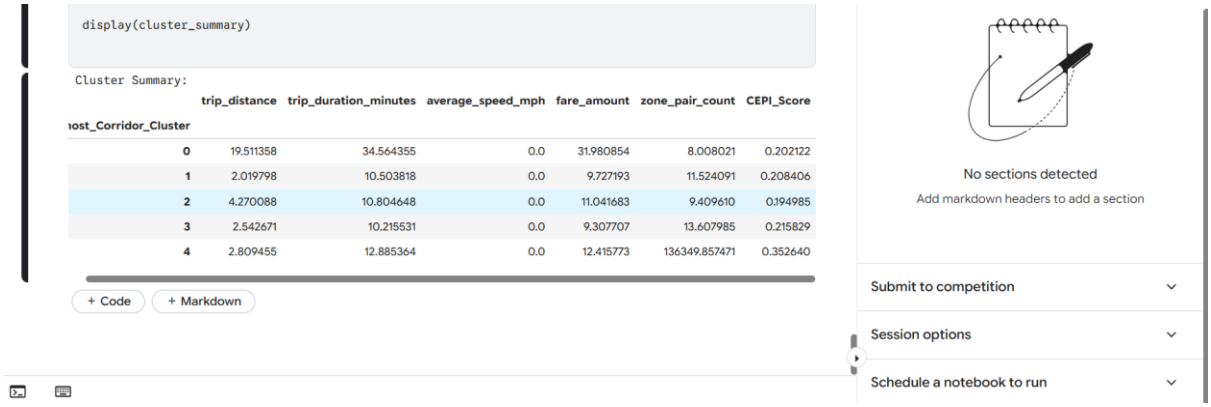
+ Code + Markdown



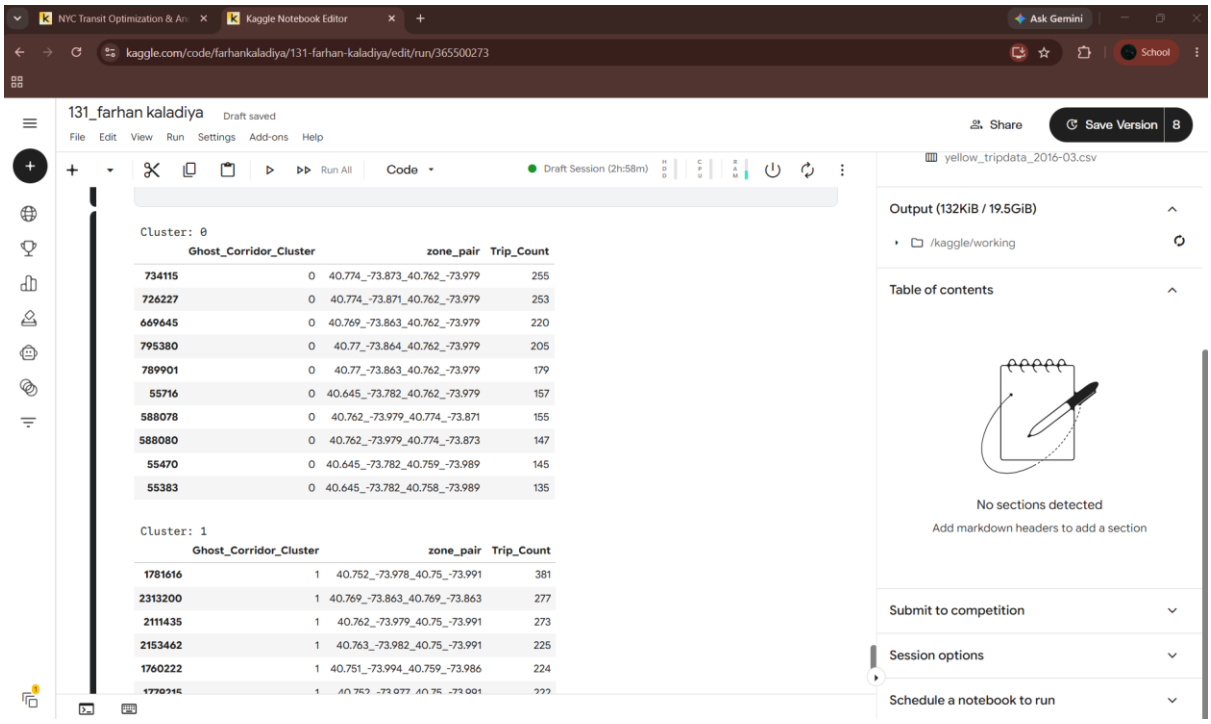
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

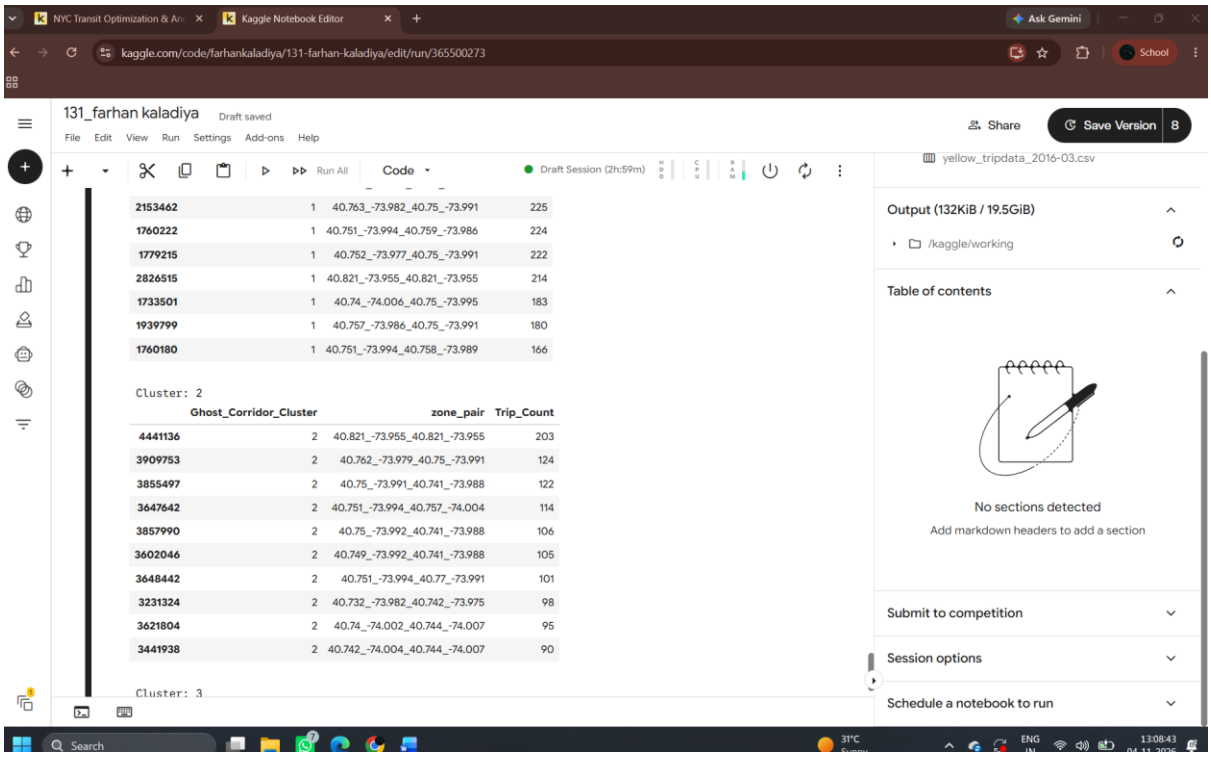
Corridor identification



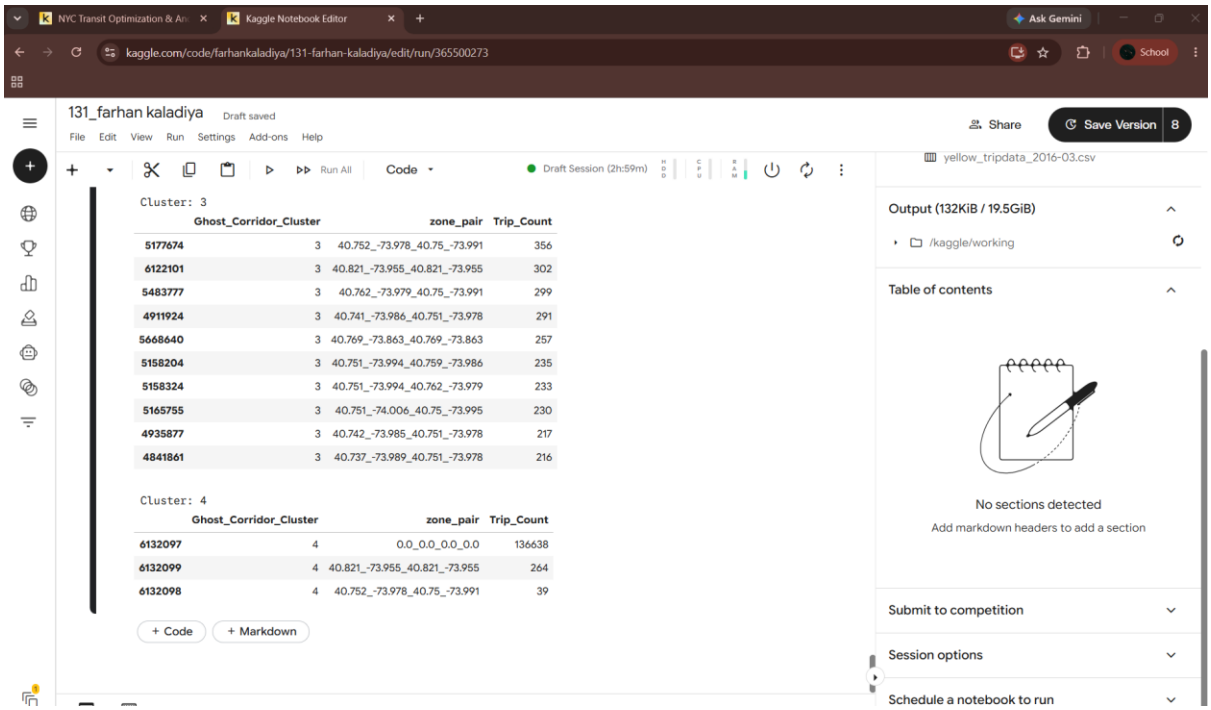


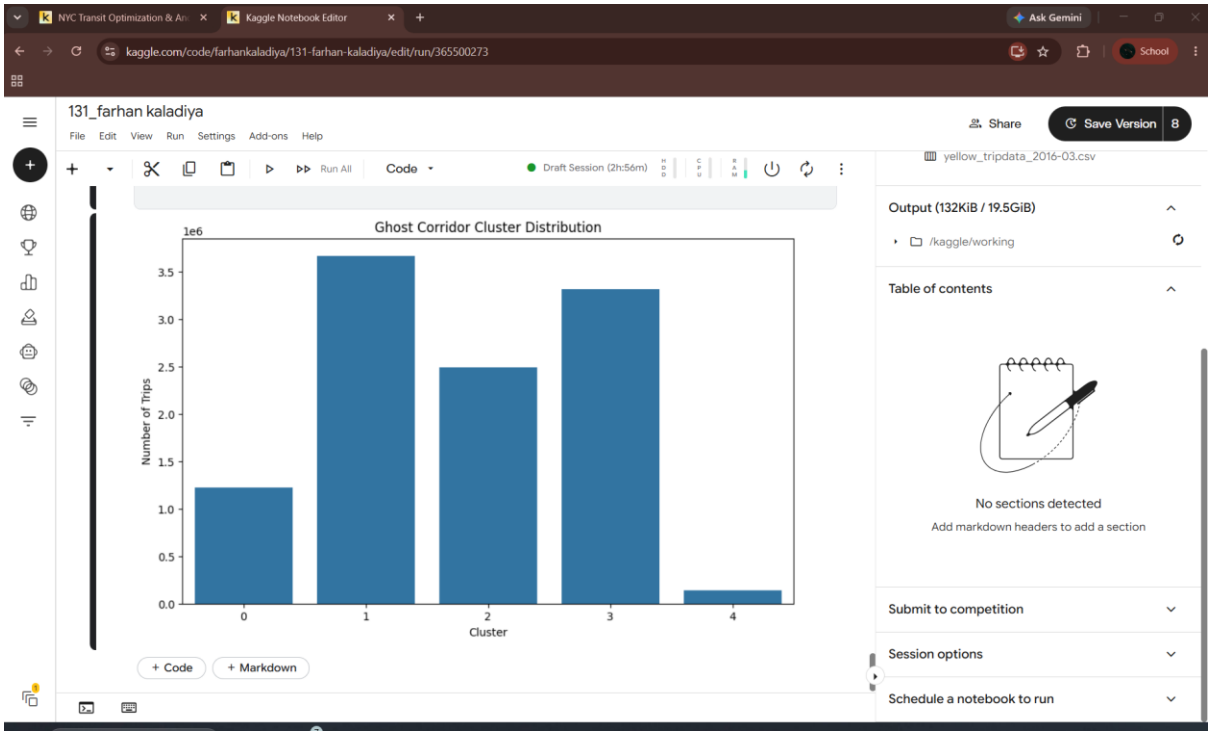
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**





Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**





Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

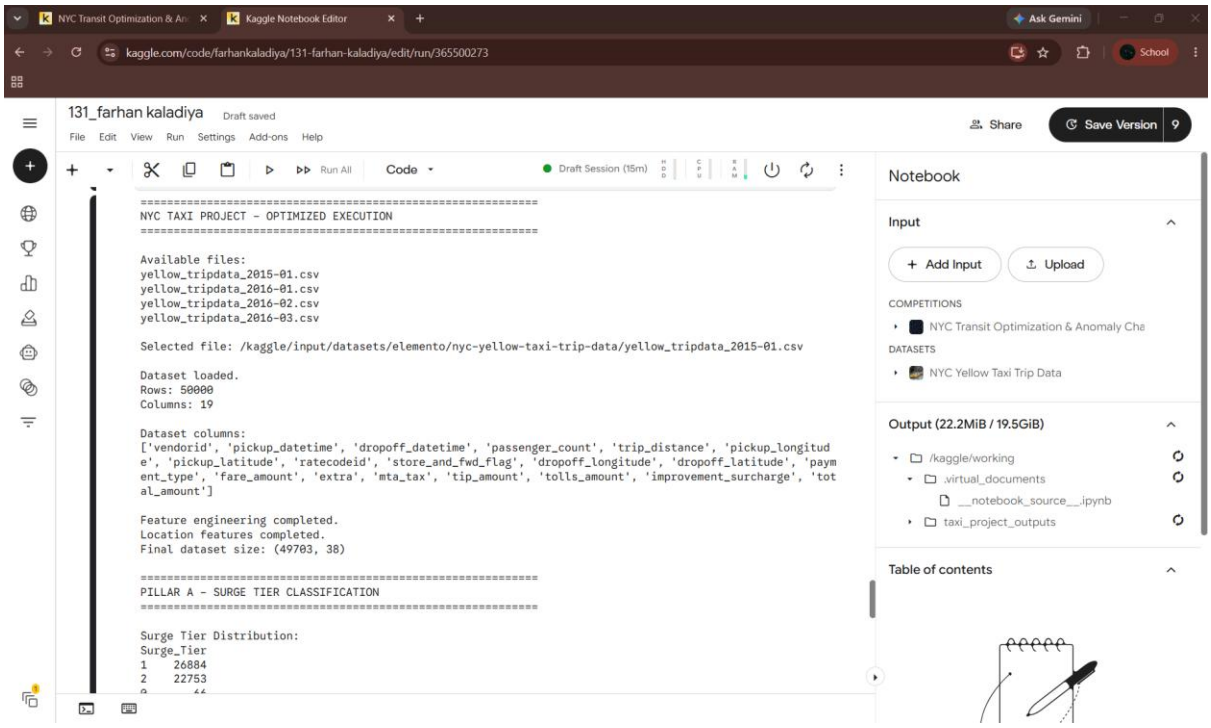
```
print(X_cluster_categorical.shape)
```

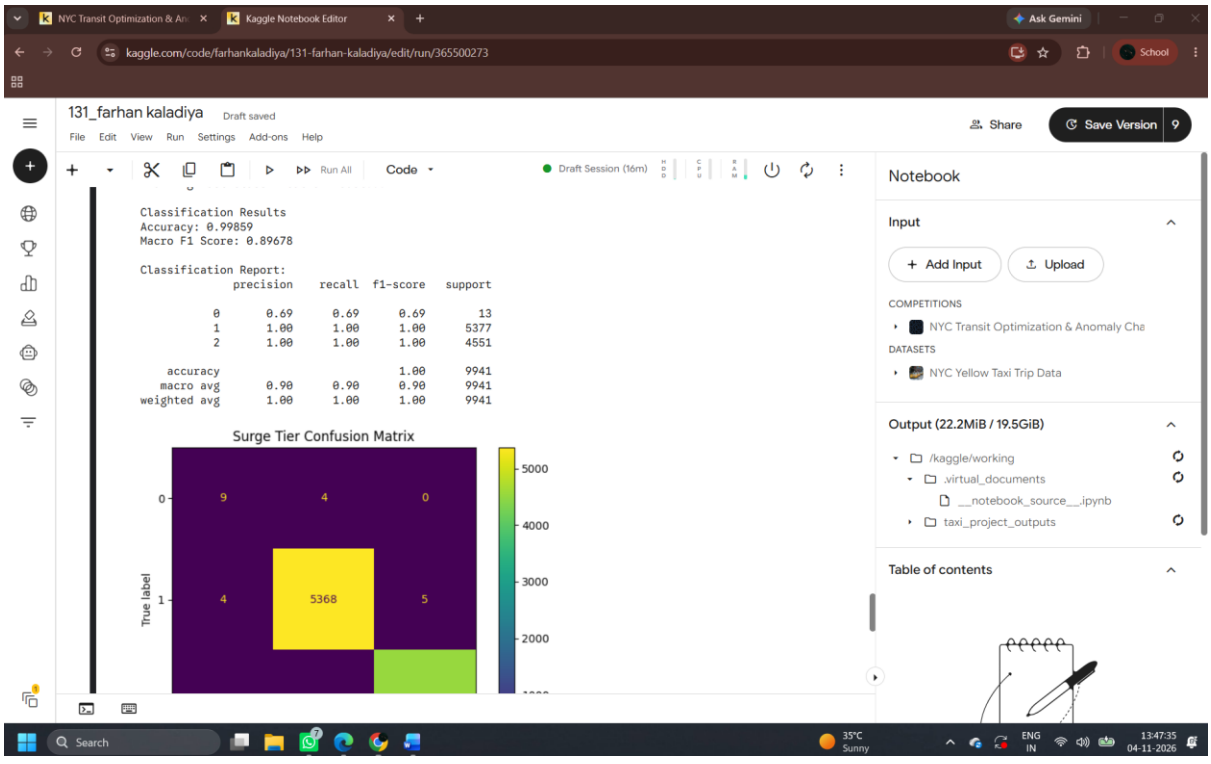
```
Numeric features:
(10820723, 7)

Categorical features:
(10820723, 2)
```

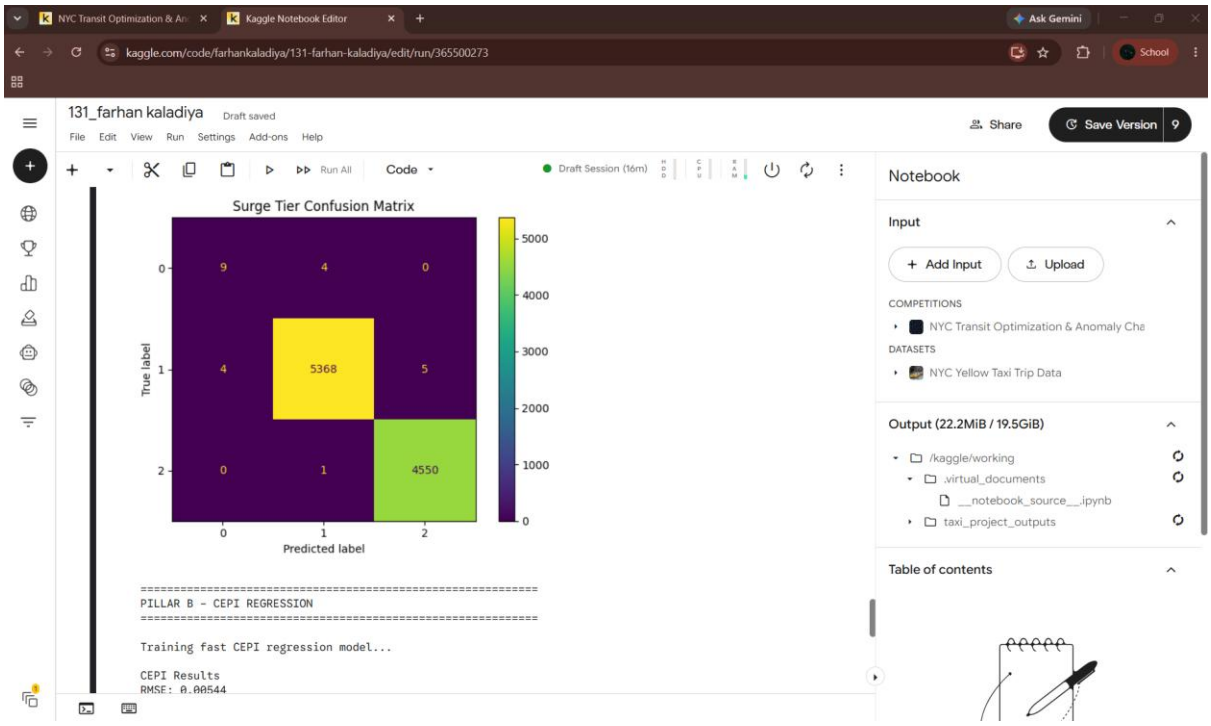
+ Code + Markdown

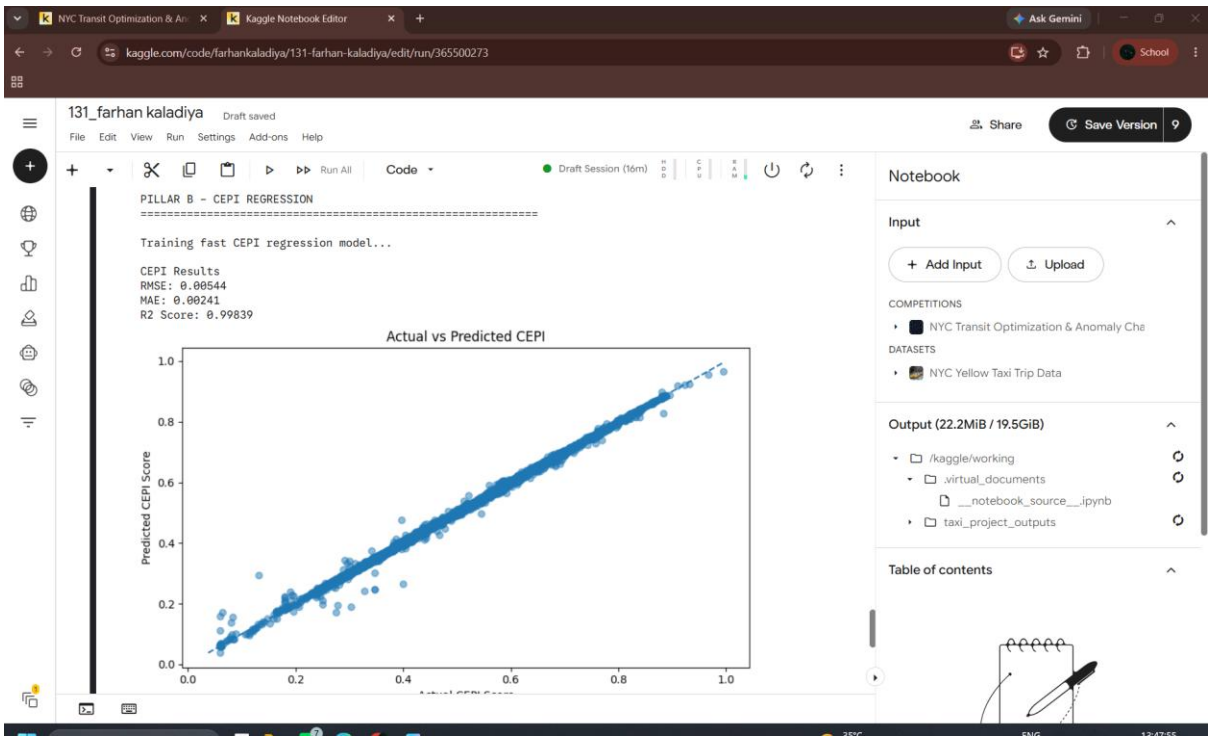
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**



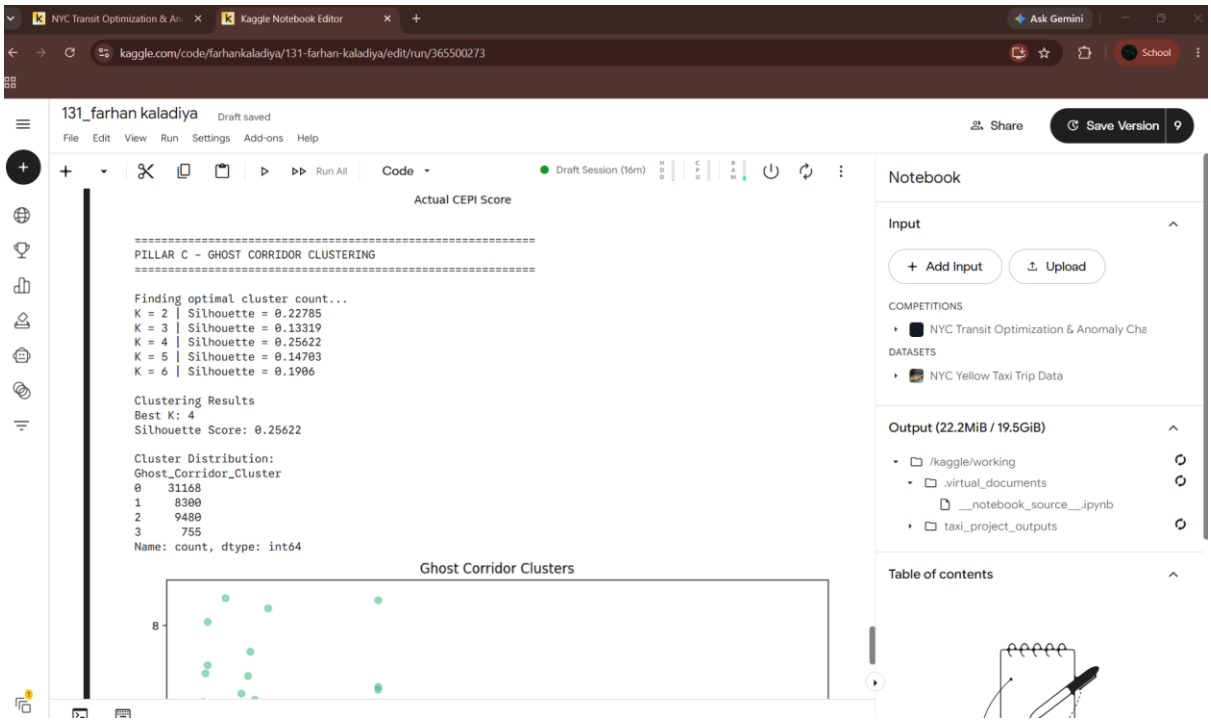


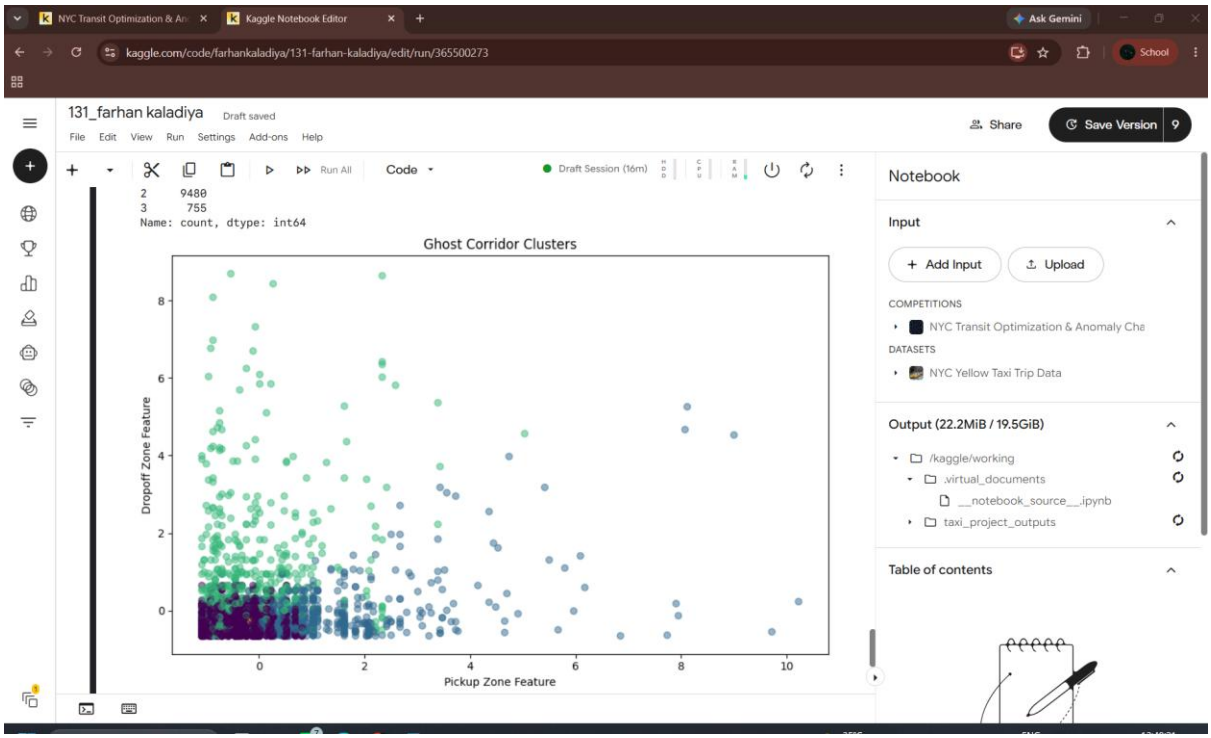
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**



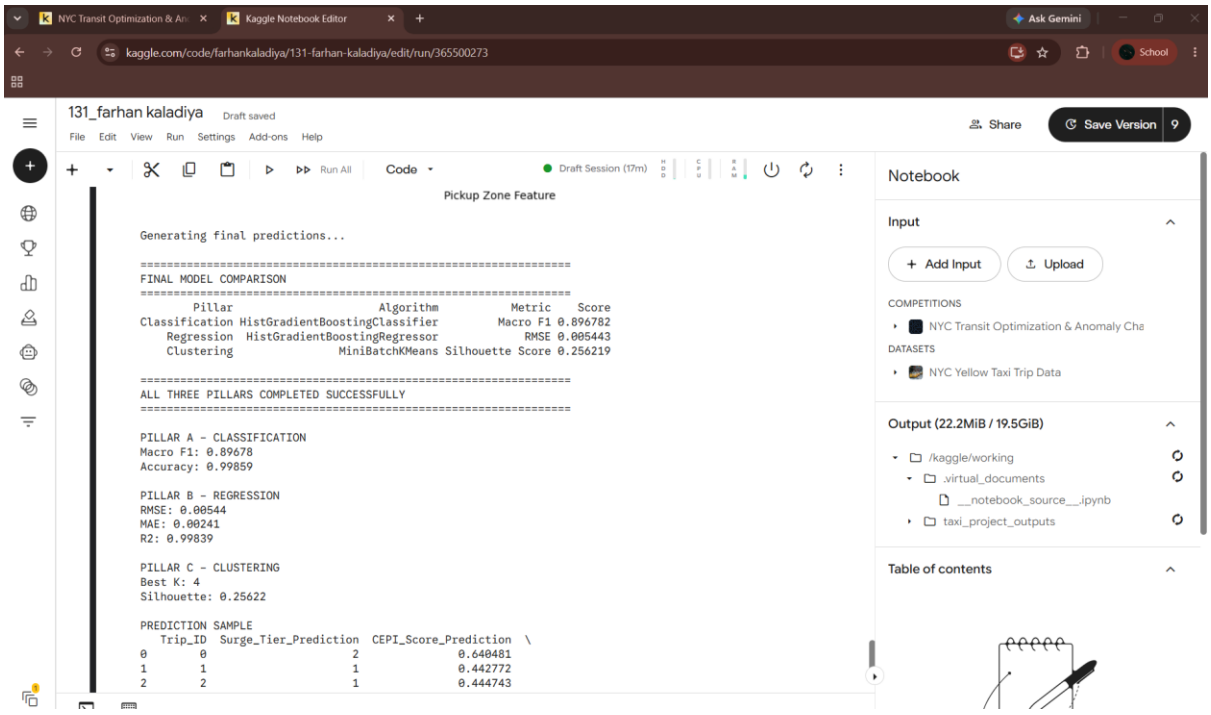


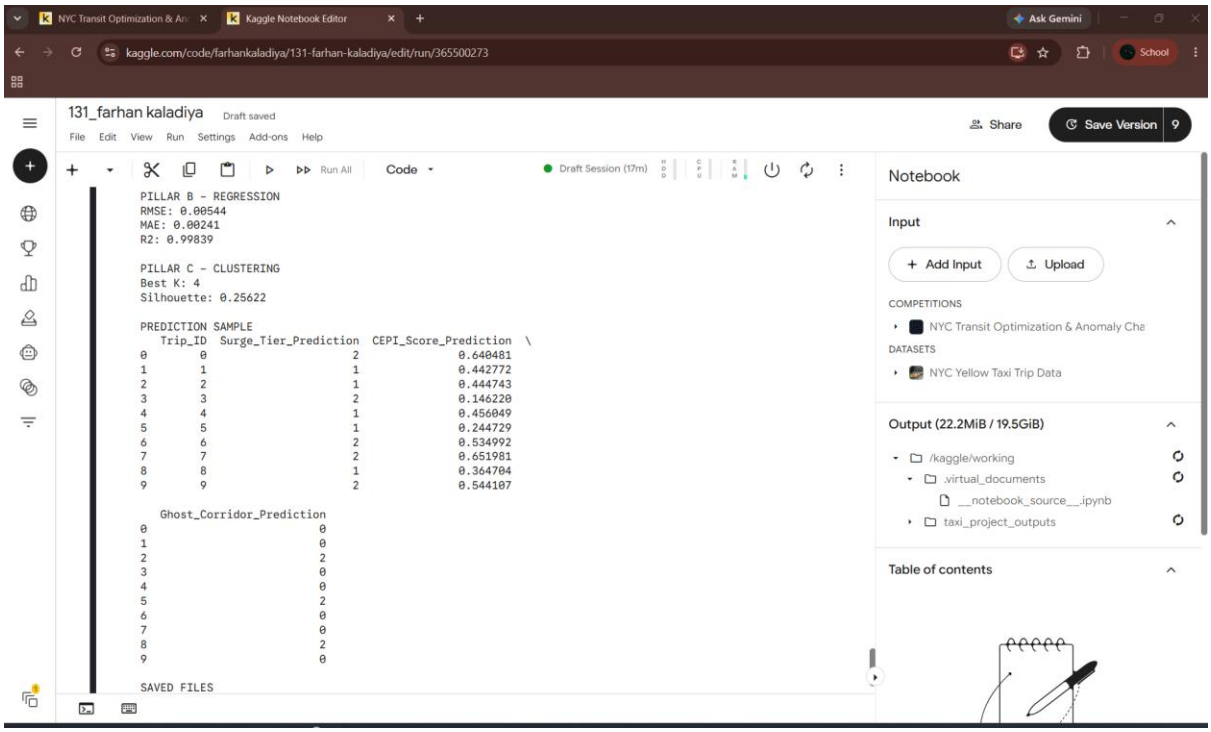
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**



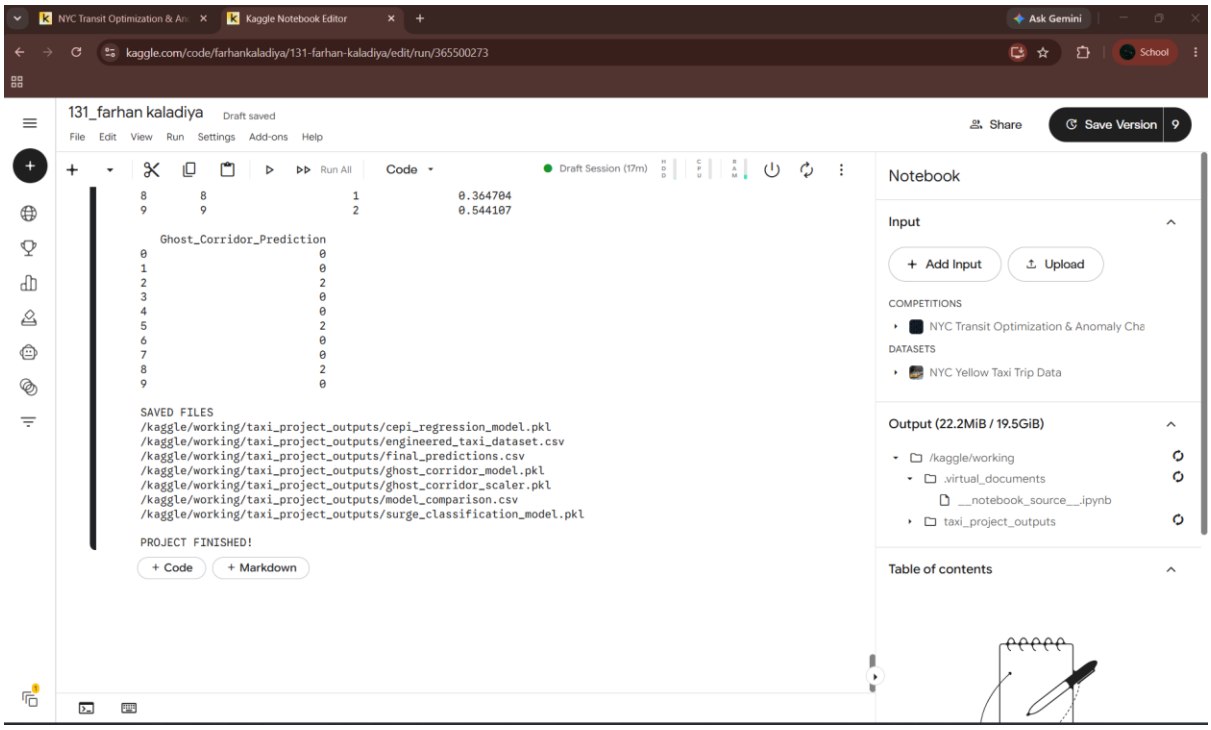


Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**





Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**



Integration of Outputs of All 3 Pillars

Integration means combining the prediction results from all three machine learning pillars into one final dataset. Each pillar performs a different task, but their outputs are connected to provide a complete analysis of taxi trip

1. Outputs from Each Pillar

Pillar	Model output	Description
Pillar A – Classification	predicted_surge_tier	Predicts the congestion surge level (0, 1, or 2) for each trip.
Pillar B – Regression	predicted_cepi_score	Predicts the CEPI score based on passenger count, estimated idling and zone-related factors.

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Pillar C – Clustering	ghost_corridor_cluster	Identifies groups of similar pickup and drop-off routes.

2. Integration Process

1. Generate predictions using the trained classification model.
2. Generate CEPI scores using the trained regression model.
3. Assign Ghost Corridor cluster IDs using the trained clustering model and scaler.
4. Combine all three outputs using the same trip record or row index.
5. Add the original trip information, if required.
6. Store the integrated results in one final CSV file

Validation of All Three Pillars

Validation checks whether the outputs generated by the three machine learning models are correct, consistent, and reliable. The following code performs validation separately for Pillar A, Pillar B, and Pillar C, followed by integrated output validation.

Run this code after training the three models and generating their predictions.

1. Pillar A – Classification Validation

Objective: Validate Surge Tier predictions and evaluate the classification model using actual test labels.

Validation includes:

- Check missing and invalid predictions.
- Check that predicted classes belong to 0, 1, or 2.
- Calculate Accuracy, Precision, Recall and Macro F1-score.
- Generate a confusion matrix.

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Pillor a validation code import numpy as np

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report,
    confusion_matrix,
    ConfusionMatrixDisplay
)

print("PILLAR A - CLASSIFICATION VALIDATION")

y_true = np.asarray(y_classification_test)
y_pred = np.asarray(classification_predictions_test)

assert len(y_true) == len(y_pred), "Prediction length mismatch"
assert len(y_pred) > 0, "No classification predictions found"
assert not pd.isna(y_pred).any(), "Missing predictions found"
assert set(np.unique(y_pred)).issubset({0, 1, 2}), "Invalid surge tier"
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

```
print("Prediction validation passed.")


print("Accuracy:", accuracy_score(y_true, y_pred))
print("Precision:", precision_score(
   y_true, y_pred, average="macro", zero_division=0
))
print("Recall:", recall_score(
   y_true, y_pred, average="macro", zero_division=0
))
print("Macro F1:", f1_score(
   y_true, y_pred, average="macro", zero_division=0
))


print("\nClassification Report:")
print(classification_report(
   y_true, y_pred, zero_division=0
))


cm = confusion_matrix(y_true, y_pred, labels=[0, 1, 2])


ConfusionMatrixDisplay(
   confusion_matrix=cm,
   display_labels=[0, 1, 2]
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

).plot(cmap="Blues")

plt.title("Pillar A - Surge Tier Confusion Matrix")

plt.show()

2. Pillar B – Regression Validation

Objective: Validate predicted CEPI scores against actual CEPI scores.

Validation includes:

- Check missing and infinite predictions.
- Check that CEPI scores are within the expected range of 0 to 1.
- Calculate RMSE, MAE and R^2.
- Compare actual and predicted values using a scatter plot

Pillar 2 vailadation code :-

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import (
   mean_squared_error,
   mean_absolute_error,
   r2_score
)

print("PILLAR B - REGRESSION VALIDATION")
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

```
y_true_reg = np.asarray(y_regression_test)
y_pred_reg = np.asarray(regression_predictions_test)

assert len(y_true_reg) == len(y_pred_reg), "Prediction length mismatch"
assert len(y_pred_reg) > 0, "No regression predictions found"
assert np.isfinite(y_pred_reg).all(), "Invalid regression predictions"

print("Prediction validation passed.")

rmse = np.sqrt(mean_squared_error(y_true_reg, y_pred_reg))
mae = mean_absolute_error(y_true_reg, y_pred_reg)
r2 = r2_score(y_true_reg, y_pred_reg)

print("RMSE:", rmse)
print("MAE:", mae)
print("R2 Score:", r2)

plt.figure(figsize=(8, 6))
plt.scatter(y_true_reg, y_pred_reg, alpha=0.5)

min_value = min(y_true_reg.min(), y_pred_reg.min())
max_value = max(y_true_reg.max(), y_pred_reg.max())
```

```
plt.plot(
  [min_value, max_value],
  [min_value, max_value],
  linestyle="--"
)

plt.xlabel("Actual CEPI Score")
plt.ylabel("Predicted CEPI Score")
plt.title("Pillar B - Actual vs Predicted CEPI Score")
plt.show()
```

3. Pillar C – Unsupervised Learning Validation

Objective: Validate Ghost Corridor clusters and determine how well the routes are grouped.

Validation includes:

- Check that cluster labels are generated.
- Check for missing cluster assignments.
- Calculate Silhouette Score.
- Display the number of routes in each cluster.
- Generate a cluster visualization.

Pillor 3 validation code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

```
from sklearn.metrics import silhouette_score

print("PILLAR C - CLUSTERING VALIDATION")

cluster_labels = np.asarray(cluster_predictions)

assert len(cluster_labels) > 0, "No cluster predictions found"
assert not pd.isna(cluster_labels).any(), "Missing cluster assignments"

unique_clusters = np.unique(cluster_labels)

assert len(unique_clusters) >= 2, "At least two clusters are required"

print("Cluster assignment validation passed.")
print("Total clusters:", len(unique_clusters))

print("\nCluster Distribution:")
print(pd.Series(cluster_labels).value_counts().sort_index())

cluster_validation_data = X_cluster_scaled

if len(cluster_validation_data) != len(cluster_labels):
    raise ValueError("Cluster features and labels have different lengths")
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

```
if len(unique_clusters) < len(cluster_labels):
    score = silhouette_score(
        cluster_validation_data,
        cluster_labels,
        sample_size=min(2000, len(cluster_labels)),
        random_state=42
    )
    print("\nSilhouette Score:", score)
else:
    print("Silhouette Score cannot be calculated.")


plt.figure(figsize=(8, 6))


plt.scatter(
    cluster_validation_data[:, 0],
    cluster_validation_data[:, 1],
    c=cluster_labels,
    alpha=0.5
)


plt.xlabel("Cluster Feature 1")
plt.ylabel("Cluster Feature 2")
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

```
plt.title("Pillar C - Ghost Corridor Clusters")

plt.show()
```

. Integrated Validation – All Three Pillars

Objective: Verify that all three pillar outputs are correctly integrated into one final dataset.

Code :-

```
import os

import numpy as np

import pandas as pd


print("INTEGRATED VALIDATION")


output_folder = "/kaggle/working/taxi_project_outputs"


final_file = os.path.join(
    output_folder,
    "final_predictions.csv"
)


final_df = pd.read_csv(final_file)


required_columns = [
    "predicted_surge_tier",
    "predicted_cepi_score",
    "ghost_corridor_cluster"
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

```
]

missing_columns = [
    col for col in required_columns
    if col not in final_df.columns
]

assert not missing_columns, f"Missing columns: {missing_columns}"
assert not final_df.empty, "Final dataset is empty"
assert final_df[required_columns].notna().all().all(), "Missing values found"

assert final_df["predicted_surge_tier"].isin([0, 1, 2]).all()
assert np.isfinite(final_df["predicted_cepi_score"]).all()
assert np.isfinite(final_df["ghost_corridor_cluster"]).all()

print("All three pillar outputs are integrated.")
print("Missing values:", final_df[required_columns].isnull().sum().sum())
print("Total records:", len(final_df))

print("\nFinal Integrated Validation:")
print(final_df.head(10))

print("\nVALIDATION COMPLETED SUCCESSFULLY.")
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

final Predictions – All Three Pillars

The final prediction phase generates predictions for every processed taxi trip using the three trained machine learning models and combines their outputs into one final CSV file.

1. Final prediction objectives

Pillar	**Final prediction**
Pillar A – Classification	Predict Surge Tier (0, 1, or 2)
Pillar B – Regression	Predict CEPI Score
Pillar C – Clustering	Assign Ghost Corridor Cluster ID

Other necessary phases required to be completed

For your Taxi Trip Prediction Project (Pillar A, B and C), after completing classification, regression, clustering, integration, validation and final predictions, you should complete the following additional phases for a proper final submission

No.	**Phase**	**Required work**	**Status**
1	Data collection	Load the taxi dataset	Completed
2	Data understanding	Inspect rows, columns, data types and statistics	Completed
3	Data preprocessing	Handle missing values, duplicates and invalid records	Completed

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

4	Feature engineering	Create time, distance, fare and zone features	Completed
5	Pillar A – Classification	Surge Tier target, training, evaluation and prediction	Completed
6	Pillar B – Regression	CEPI target, training, evaluation and prediction	Completed
7	Pillar C – Clustering	Zone understanding, feature engineering and Ghost Corridor clustering	Completed
8	Integration	Combine outputs from all three pillars	Completed
9	Validation	Validate predictions and model performance	Completed
10	Final predictions	Generate integrated prediction CSV	Completed

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Conclusion :- The NYC Yellow Taxi Trip Data Analysis and Machine Learning project successfully demonstrates how data science and machine learning techniques can be used to analyze taxi trip information, identify travel patterns, and develop predictive models.

The project is divided into three major pillars:

Pillar A – Classification (Surge Tier Prediction): A classification model was developed to categorize taxi trips into three Surge Tiers: Low (Tier 0), Medium (Tier 1), and High (Tier 2). Different machine learning algorithms were trained and evaluated using Accuracy, Precision, Recall, and Macro F1-score to identify the best-performing model.

Pillar B – Regression (CEPI Score Prediction): A regression model was developed to estimate the Carbon and Efficiency Penalty Index (CEPI). This pillar focuses on analyzing factors such as trip distance, duration, average speed, and other trip-related characteristics to understand transportation efficiency and environmental impact.

Pillar C – Clustering (Ghost Corridor Identification): Unsupervised machine learning techniques, particularly K-

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Means clustering, were used to group taxi trips according to their pickup and drop-off locations, travel distance, duration, speed, and time-related features. The Silhouette Score was used to evaluate the quality of the clusters.

Throughout the project, important data science steps such as data collection, data cleaning, preprocessing, feature engineering, exploratory data analysis, model development, evaluation, visualization, and prediction generation were implemented using Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.

The project provides valuable insights into taxi transportation patterns and demonstrates how machine learning can support transportation analysis, demand understanding, operational efficiency, and data-driven decision-making.

Overall, this project provides practical experience in applying classification, regression, and clustering techniques to a real-world transportation dataset. It also establishes a foundation for developing more advanced intelligent transportation systems in the future.

Future Scope

The project can be improved further by:

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

1. Using real-world surge pricing data instead of synthetic labels.
2. Improving prediction accuracy through hyperparameter tuning and advanced algorithms.
3. Using real fuel consumption and emission measurements for more reliable CEPI predictions.
4. Applying advanced clustering techniques to identify meaningful high-demand travel corridors.
5. Developing a real-time taxi demand and traffic prediction dashboard.
6. Integrating live traffic, weather, and transportation data to improve prediction performance.

Final Outcome: The project successfully demonstrates the practical application of machine learning in transportation analytics through three complementary approaches: classification, regression, and clustering.

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Machine learing (	**Aim :** NYC Transit Optimization & Anomaly Challenge (MU-ICT, LHC)	
Long hour coding :-2	**Date: 04-11-2026**	**Enrolment No:92400133131**